NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES

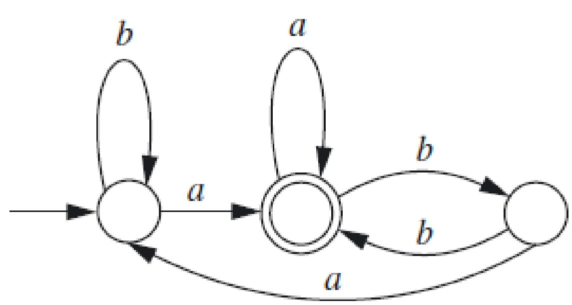
Theory of Automata (I)
Final Examination, Fall 2012
Section A,B,C,D,E

Total Marks: 100 (20 objective+80 subjective)
Time: 3 Hours

Subjective Part: (2.5 hrs)

Q1. Regular languages
(i) [5 points]
Develop DFA for the language over the alphabet {a, b, c} that accepts strings whose last two characters are the reverse of first two characters.

(ii) [10 points]
Use the state elimination method to convert the following DFA to a regular expression. You must show all the intermediate steps.



Q2. Context Free Languages
(i) [10 marks]
Build CFG for the following language: $\{a^i b^j c^k \mid j \text{ not equal to } i + k\}$

(ii) [5 marks]
Is the language given below a CFL? If not prove via Pumping Lemma.
L = {www| w belongs to {0,1}*}

(iii) [10 marks]
Constructing an LL (1) parser.
Consider the following CFG over ∑ = { ⊕ , ∧, d}
S → S⊕S | S∧S | d
Convert it into LL (1) Grammar and then build its LL (1) parser.

Q3. Turing Machine

Note:

- Give short description of your TM in few points.
- In case you are using any blocks (composite TM) to design this Turing machine, you need to specify the internal design of those blocks as well.
- **For Section A,B,C,D design TM for the parts below & for Section E design Deterministic TM for the parts below.**

(i) [10 marks]

Design a generic 1-Tape Turing machine "EQUAL" that will accept the input if it contains an equal number of 0's and 1's. Input is placed on the tape. The machine should accept 01111000, 11110000, 10110100…..

(ii) [15 marks]

Design a generic 3-Tape Turing machine for Div (a^m, a^n) function where m and n > 0. You need to divide string a^m with string a^n . Initially Tape-1 contains a^m and then a^n separated by # and tape 2 & 3 are blank. After division, tape-3 should contain the quotient.

Sample Run:

Input: (on tape-1 initially)

#	a	a	a	a	a	#	a	a	#	

Output: (on tape-3 after execution)

#	a	a	

(iii) [15 marks]

Design a standard 2-tape Turing machine with alphabet set Σ = {a, b, c} that sorts the symbols in the input string in the lexicographic order such that in the output string all a's appear before the b's and c's, and all b's appear before the c's (For example, on input bbcacb, the output would be abbbcc.).

You can assume that the input is on tape 1. Your output should be on tape 2 & input on tape 1 when the TM halts.

Sample Run:

Input(on tape-1 initially)

Δ	b	b	c	a	c	b	Δ	Δ	Δ

Output: (on tape-2 after execution)

Δ	a	b	b	b	c	c	Δ	Δ	Δ

Theory of Automata

Final Exam (Sec A, B), Spring 2016

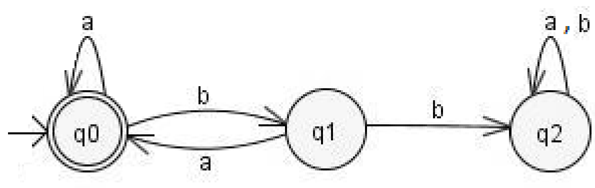
Roll Number: ___________ Section:_______

Date: 13-5-2016 **Marks: 85** **Time: 180 mins.**

You have to provide answer on this question paper. Answer on extra sheet will not be marked. You are allowed to take extra sheets for rough work. But that extra sheet will not be collected.

Problem 1: Arden's Lemma **[5 + 10 points]**

(a) Derive regular expression of the following DFA using Arden's Lemma.



Solution:

$q_0 = q_0a + q_1a + \wedge \rightarrow (1)$

$q_1 = q_0b \rightarrow (2)$

$q_2 = q_1b + q_2(a+b)$

put (2) in (1)

$q_0 = q_0a + q_0\,ba + \wedge$

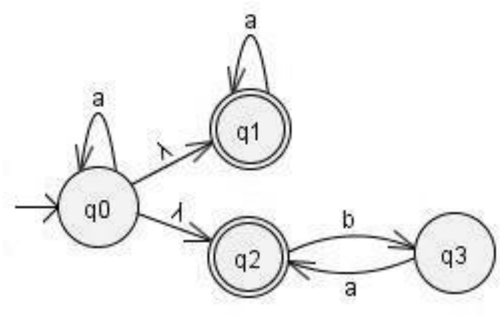
$q_0 = q_0a + q_0\,ba + \wedge$

$q_0 = q_0(a + ba) + \wedge$

Applying adren lemma

$q_0 = (a + ba)^*$

(b) Convert following NFA-null to DFA



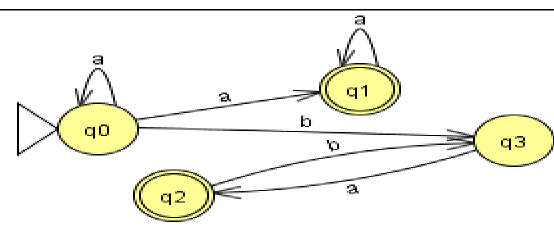
Solution:

state	null	a	b
q0	{q0,q1,q2}	{q0}	Φ
q1(Final)	{q1}	{q1}	Φ
q2(Final)	{q2}	Φ	{q3}
q3	{q3}	{q2}	Φ

Given NFA – null table:

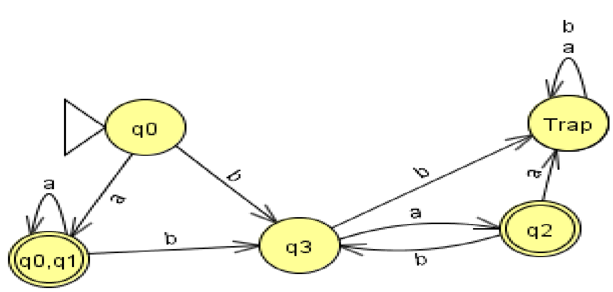
Covert to NFA

state	a	b
q0	{q0,q1}	{q3}
q1(Final)	{q1}	Φ
q2(Final)	Φ	{q3}
q3	{q2}	Φ



Convert to DFA

state	a	b
q0	{q0,q1}	q3
q2(Final)	trap	q3
q3	q2	trap
{q0,q1}(Final)	{q0,q1}	q3
trap	trap	trap



Problem 2: Parser **[10 + 5 points]**

a. Apply the CYK algorithm on the given CFG to determine whether this string 'bbaba' is accepted or not.

S→ AB
A→ BX | a
B →CC | b
C→ AB | a
X→b

Solution:

S,C bbaba				
- bbab	- baba			
- bba	- bab	B aba		
A bb	- ba	S,C ab	- ba	
B,X b	B,X b	A,C a	B,X b	A,C a

b. Prove whether the following grammar is ambiguous or not. If grammar is ambiguous then remove the ambiguity.

S→ a S b | ab | T | ^
T→ aT | ^

Solution:
S→aSb | T
T→ aT| ^

Roll Number: ___________ Section:_______

Date: 13-5-2016 **Marks: 85** **Time: 180 mins.**

Problem 3: CFL **[5 + 10 points]**

(a) Remove Null productions from the following grammar.
Tell whether the resulting grammar is CNF or not?

S→ AB | ^

A→ aAb | Ba |^

B→bAb |aaB | ^

Solution:
S→A|B|AB| ^
A→aAb|Ba|ab|a
B→bAb|aaB|bb|aa

Not in CNF

(b) Develop PDA for the following language

$$L = \{x \mid x \in \{a, b, c\}^* \;\&\; n_c(x) = 2\,[n_a(x) + n_b(x)]\}$$

Where,

$n_a(x)$ = number of a's in x

$n_b(x)$ = number of b's in x

$n_c(x)$ = number of c's in x

Problem 4: Pumping Lemma **[10 points]**

Show that the following language is not context free using pumping lemma.

$$\mathbf{L = \{x \mid x \in \{a, b, c\}^* \ \&\ n_a(x) = 2\, n_b(x) \ \&\ n_c(x) = n_a(x) + n_b(x)\ \}}$$

Where,

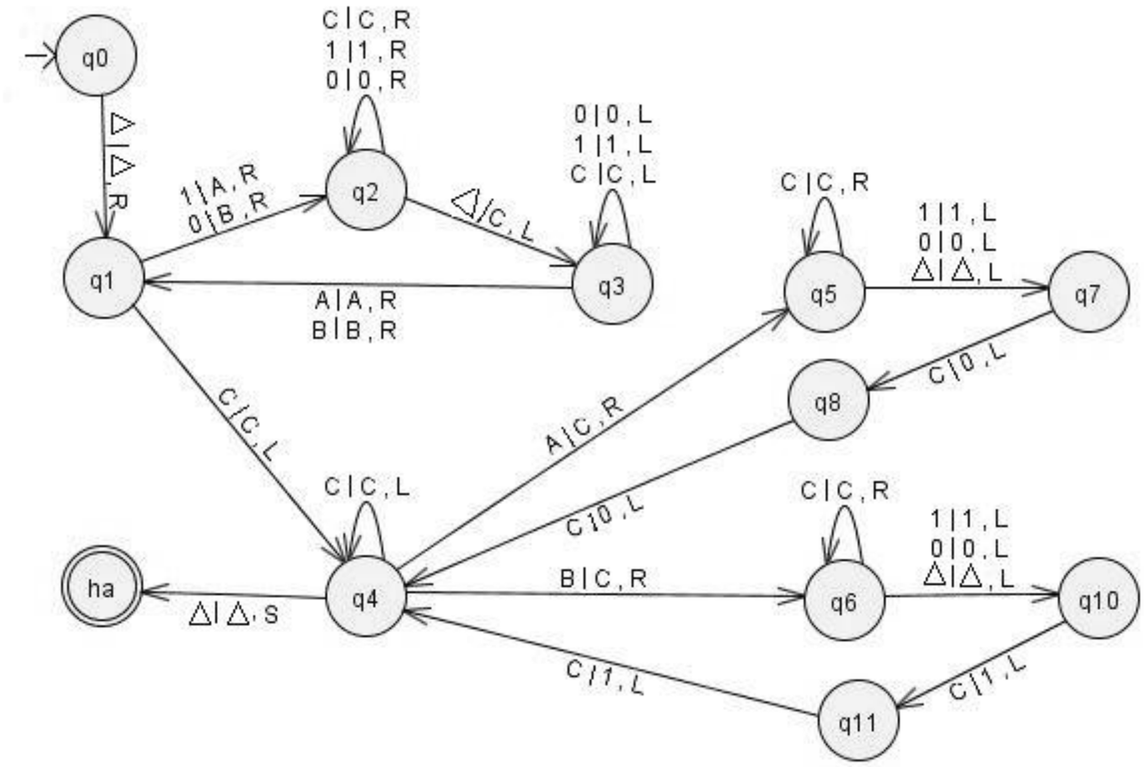
$\mathbf{n_a(x)}$ **= number of a's in x**
$\mathbf{n_b(x)}$ **= number of b's in x**
$\mathbf{n_c(x)}$ **= number of c's in x**

Roll Number: ___________ Section:_______

Date: 13-5-2016 **Marks: 85** **Time: 180 mins.**

Problem 5: Single Tape Turing Machine **[15 points]**

You are given with a single tape TM.



Output on given string: You have to run the TM on the string 101101 (make sure input is enclosed between Δ's on TM). After processing what is the output of tape?

Tape you are given:

Δ	1	0	1	1	0	1	Δ	Δ	Δ

Output Tape:

Δ	0	0	1	1	0	0	0	0	1	1	0	0	Δ

Description of TM: Tell in words what it is doing (answer should be generic).

Problem 6: MultiTape Turing Machine **[15 points]**

The 3 tape Turing Machine **Equal** operates as follows:

TM halts in ***halt state*** with ***stationary head*** if it accepts the following language.

$$L = \{x \mid x \in \{a, b, c\}^* \ \& \ n_c(x) = 2\,[n_a(x) + n_b(x)]\}$$

Where,

$n_a(x)$ = number of a's in x

$n_b(x)$ = number of b's in x

$n_c(x)$ = number of c's in x

National University of Computer and Emerging Sciences, Lahore Campus

Course:	Theory of Automata	Course Code:	CS 301
Program:	BS(Computer Science)	Semester:	Fall 2016
Duration:	180 Minutes	Total Marks:	55
Paper Date:	28-DEC-16	Page(s):	9
Section:	A,B,C,D	Section:	
Exam:	Final	Roll No:	

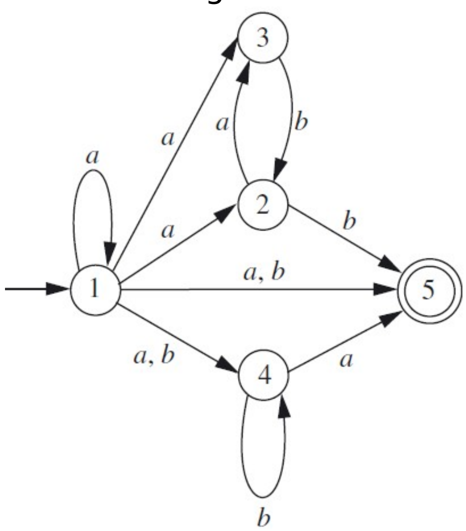
Instruction/Notes:

1. All students can bring **one hand written A4 help sheet**in the exam hall
2. Sharing of A4 sheet is strictly **NOT ALLOWED**
3. Answer in the space provided
4. You can ask for rough sheets but they will not be graded or marked

Good luck!

QUESTION 1 (Marks:5)

Convert the given NFA to DFA

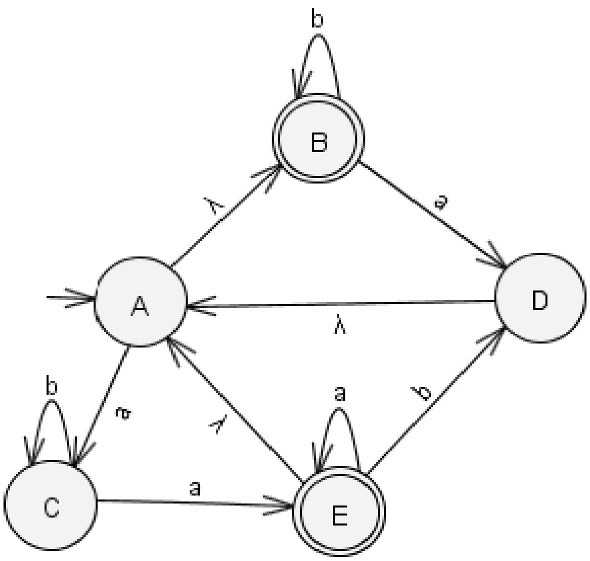


Solution:

state	a	b
1	12345	45
12345	12345	245
45	5	4
245	35	45
5	Trap	Trap
4	5	4
35	trap	2
2	3	5
3	trap	2
Trap	Trap	tra

QUESTION 2 **(Marks:1 +4)**

Given the NFA-null



a. Using the extended transition function of an NFA-null find out if the string ***abab*** is part of the language or not.

b. Convert the above NFA-null to NFA

Solution:

a. {A,B,D}

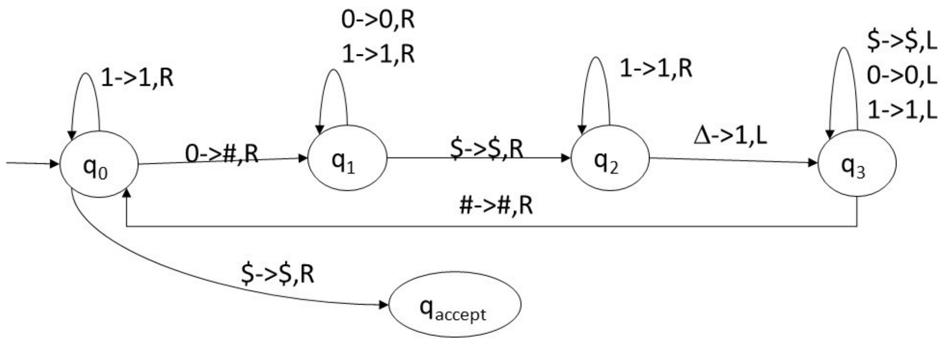
b.

State	a	b
A	{A,B,C,D}	{B}
B	{A,B,D}	{B}
C	{A,B,E}	{C}
D	{A,B,C,D}	{B}
E	{A,B,C,D,E}	{A,B,D}

QUESTION 3 (Marks: 1+1+1+1+2)

You are given a single tape TM.Blanks are represented by Δ. The edges are represented by the notation:

Current Symbol->Replaced symbol, Direction



a. You have to run the TM on the following strings and write the output on the tape when the machine halts.

Input on tape	**Output on tape**
101$	**1#1$1**
100010$	**1###1#$1111**
111$	**111$**
0000$	**####$1111**

b. What does the above machine do? (answer should be generic).

It count number of 0's in the given input

QUESTION 4 (Marks: 1+4)

Suppose we are given three strings ϵ {a,b,c}* as input, we have to determine the index of the string which has maximum length.

Example: if String1=abd String2=cccc and String3=cbabbb are the input, the output= 3

a. Write an algorithm, in plain English, to solve this problem using a TM.

b. Create a single tape deterministic Turing machine for your algorithm. Initially assume that all three input strings are separated by the delimiter #. Also, the input starts with # symbol. So when given the input on tape as:

#abd#cccc#cbabbb#, the tape when TM halts should be: #abd#cccc#cbabbb#3

Idea: Compare String 1 and 2, whichever string is larger in this comparison compare it with 3, which ever string is larger in this comparison write its index on end of tape. If in any comparison both strings are of equal length, one will be selected randomly.

Algorithm:

Step 1: Look for 1st uncrossed symbol of String 1,
If found cross it go to step 2
If not found, go to step 3 // string 2 is larger or equal than string 1

Step 2: Look for 1st uncrossed symbol in String 2,
If found cross it and go to step1
If not found, go to step 4 // string 1 is larger or equal than string 2

Step 3: // string 2 is larger or equal than string 1.
Uncross all the symbols of String 2//restore string 2 compare with String 3

Step 3b: Look for 1st uncrossed symbol in String 2,
If found cross it and go to step3c
If not found; go to end of tape and write 3, halt. // string 3 is greater of equal to string 2 and 1

Step 3c: Look for 1st uncrossed symbol in String 3,
If found cross it and go to step3b
If not found go to end of tape and write 2, halt. // string 2 is greater of equal to string 3 and 1

Step 4: // string 1 is larger or equal than string 2.
Uncross all the symbols of String 1 //restore string 1 compare with String 3

Step 4b: Look for 1st uncrossed symbol in String 1,
If found cross it and go to step4c
If not found; go to end of tape and write 3, halt. // string 3 is greater of equal to string 1 and 2

Step 4c: Look for 1st uncrossed symbol in String 3,
If found cross it and go to step4b
If not found go to end of tape and write 1, halt. // string 1 is greater of equal to string 3 and 2

QUESTION 5 (Marks: 3+3+3+3)

a. Using induction prove that any string $s \in \{0,1\}^*$ can be written as $s = xy$ such that $n_0(x) = n_1(y)$, where $n_a(z)$ denotes the number of occurrences of the symbol a in the string z. $|x| >= 0$ and $|y| >= 0$

Step 1: Base Case: for s= null
X=null y =null
S= XY=null
$N_0(X)=N_1(Y)$

Step 2: Let the given statement be true for s'
|s'|=k
X=0 , Y=1

$S'=X^kY^k = 0^k1^k$
Such the n0(X)=n1(Y)

Step 3: For s'' of length K+1
If
$S''= X^{k+1} Y^{k+1} = 0^{k+1} 1^{k+1}$
$S''= XX^kY^kY = 00^k1^k1$
S''= X S'Y = 0 S'1
S'' = S' as |X|=|0|=|Y|=|1|=1

Proved n0(X)=n1(Y)

Is the following statement correct for any two languages L_1 and L_2? Give a proof:

$(L_1L_2)^* = L_1^*L_2^*$

Dis-proof by contradiction:

Let L1= {0}
Let L2= {1}

{L1L2}*= {01}*= {null, 01,0101,010101,….}
L1*={null,0,00,000,….}
L2*= {null, 1, 11, 111,….}
L1*.L2*= {null, 01,001,0011,….}

0101 is in $(L_1L_2)^*$ but not in $L_1^*L_2^*$ so $(L_1L_2)^*$!= $L_1^*L_2^*$

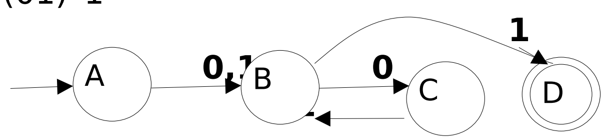
b. Give a regular expression for all binary strings so that there is an even number of 0's between any two 1's, e.g., the strings 10010000111001 and 110010000001 belong to the language. (you can consider zero as an even number). The strings 101 and 10001 do not belong to the language.

RE: (1* + 0* + 1(00)*1)*

c. Convert the given regular expression into an NFA-null

(0+1)(01)*1

Solution:



QUESTION 6 (Marks: 4+4)

a. Convert the following CFG to a PDA:

S ->aB | bA
B -> Cab | ε
C ->aa | bb | aaC | abC | ε
A -> aba | aab | ε

Solution:

State	Input	Top of Stack	Move
Q1	-	Z_0	Q2,Z_0S
Q2	-	S	(Q2,aB), (Q2,bA)
Q2	-	A	(Q2,aba), (Q2,aab), (Q2,ε)
Q2	-	B	(Q2,Cab), (Q2, ε)
Q2	-	C	(Q2,aa),(Q2,bb) , (Q2,aaC) ,(Q2, abC) ,(Q2,ε)
Q2	A	a	Q2,pop
Q2	-	Z_0	Q3, Z_0

b. Make a deterministic PDA for the following language L.

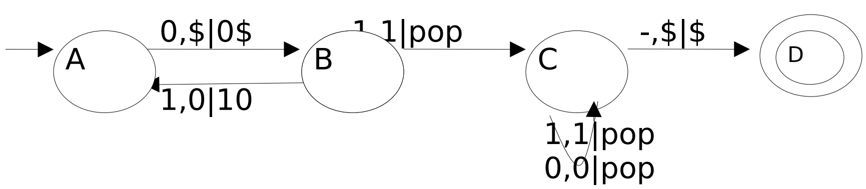
L = {$(01)^n(10)^n$ | n> 0}

Assume that the initial stack has a $ placed on it. You have to represent each edge with one of the following notations BUT NOT BOTH:

input,stack_top_character / string_that_replaces_stack_top

OR

input, pop character x --> push character y



QUESTION 7 (Marks: 3+5+3+3)

a. Show that if any language L is decidable then the complement of that language L’ is also decidable

If the language L is decidable then there must be some Turing machine M that will decide it, that is for every input x it will either accept or reject.
Using M we can create a decider N of L’ as well that will work as follow.

N= “For input x

1. Simulate M for x
2. If M accepts, reject
3. If M rejects accept

b. Given that L_1 is undecidable and Turing recognizable:
L_1: (B | B is an ambiguous CFL)
Describe an algorithm for recognizing L_1
Now given that if a language is undecidable, its compliment is Turing Unrecognizable. **Is L_2 decidable, undecidable or unrecognizable?**
L_2: (B | B is an unambiguous CFL)

If L1 is Turing recognizable then there is a Turing Machine M that can recognize it. Following is the description of M

M: “Input <B> where B is CFL
For every x ϵ ∑* (terminals)
Find every possible parse tree to derive x from <B>
If there are more than 2 parse trees of x; accept “
As we can see that there are infinite number of possible x , therefore reject state will nevere be reached. This makes M a recognizer of L1

L2 is unrecognizable:
Proof by contradiction:
Let L2 be recognizable. As L2 is complement of L1, if both L1 and it complement are recognizable then L1 is decidable. This is a contradiction from the given statement that L1 is undecidable. Therefore our assumption that L2 is recognizable is false, making L2 unrecognizable.

c. Euclidean Algorithm to decide if the Greatest Common Divisor (GCD) of two numbers a and b is c, i.e, decide if GCD(a,b) = c

```
Step 1: if a=b goto step 3
Step 2: if a>b
            a:=a-b
    else
            b:=b-a
    goto step1
Step 3:  GCD= a
    if (a==c) accept else reject
```

Does the above algorithm belong to P class? Give reason

Yes, the given algorithm is in P class as the maximum number of iterations will be N= max (a,b). The algorithm can be simulated in Deterministic Turing machine in polynomial time.

Does the above algorithm belong to NP class? Give reason.

d. We have a set X of n numbers. Determine if X has a subset, such that the GCD of all numbers in that subset is 6. So for example if X= {1 3 4 6 8 10 20 18 36 42 48 60}, {6, 8, 60} is subset that has GCD 6. And X= {1, 2, 3, 4, 5, 7, 8, 13} has no such subset.

Does the above algorithm belong to P class? Give reason. Does the above algorithm belong to NP class? Give reason.

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Course:	Theory of Automata	Course Code:	CS-2207
Program:	BS(Computer Science)	Semester:	Fall 2017
Duration:	3 Hours	Total Marks:	70
Paper Date:	11-12-17	Weight	45%
Section:	A,B,C,D,E,F	Page(s):	10
		Roll. No:	
Exam:	Final Exam	Section:	

Instruction/Notes:
- All the questions are to be attempted on this question paper
- You can use rough sheet but answers and working should be shown on this question paper.
- You must show working for all the questions, no marks will be given without proper working.
- Please don't attach any extra sheet with the question paper.

Question 1: Part (a) [marks 10]

<u>Construct A Deterministic Finite Automata</u>:

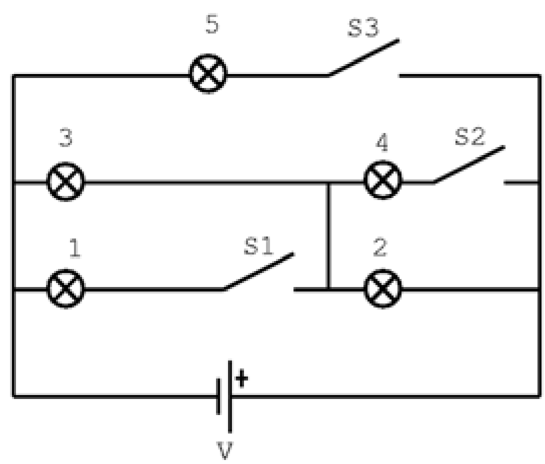


Figure 1A

Above diagram shows a circuit having three pushbutton switches and 5 bulbs. Each bulb is turned on if its corresponding circuit is closed. You are to create DFA of the above circuit. The states of the DFA should reflect the state switches (i.e. on or off). Initial state of your DFA should be such that all bulbs are off, and final state should be such that only bulb 3 and 2 are turned on. Input of your DFA will belong to {S1, S2, S3}*.

Sample input: S1 S2 S2 S3 S1 S2

Which means first S1 is pressed then two times S2 is pressed then S3, S1 and finally S2 is pressed. On each press of a switch its state will be toggled. i.e. if switch was off it will be turned on and vice versa.

NOTE: consider S1 as one symbol, same for S2 and S3 as well.

Question 1: Part (b) [marks 10]

Minimal Automata

Minimize the DFA given in figure 1B, draw state diagram of your final DFA. Also show your working.

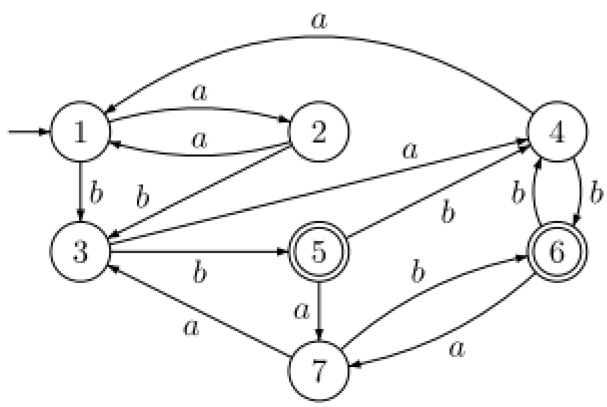


Figure 1B

2						
3						
4						
5						
6						
7						
	1	2	3	4	5	6

Question 2: Part (a) [marks 10]

Context Free Grammar

Construct CFG (Context free grammar) of L_1 where L_1 is context free language.

L_1: { $a^n b^m c^k$| n>=m and k=n-m}

Question 2: Part (b) [marks 10]

<u>Top down deterministic Parser</u>

Construct a Top down deterministic Parser for the following CFG.

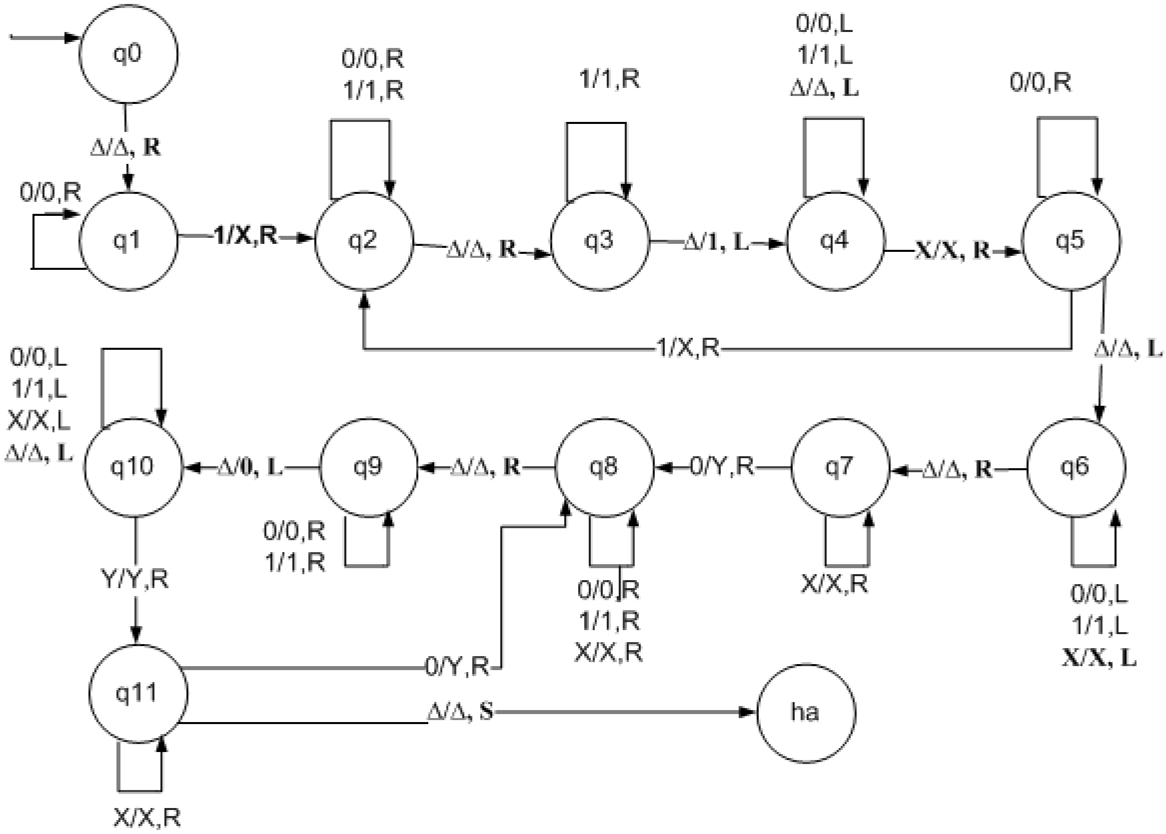
S→ A#

A→aAbA| null

Note: # is same as $

Question 3: Part (a) [marks 10]

You are given with a single tape TM.



Output on given string: You have to run the TM on the string 111010101 (make sure input is enclosed between Δ's on TM).After processing what is the output of tape? (Show working)

Tape you are given:

Δ	1	1	1	0	1	0	1	0	1	Δ

Output Tape:

Description of TM: Briefly explain in words what it is doing (answer should be generic).

Question 3: Part (b) [marks 20]

Turing Machine

Given 2 Strings X and Y such that X,Y ϵ $\{a,b\}^+$

Construct a Turing Machine that outputs the number of times Y appears in X.

Your Turing machine will have 3 tapes. Tape 1 will have X as input, Tape 2 will have Y as input, and on Tape 3 you have to write the output. As your output is the number of times Y appears in X, it should be in unary form.

Following is an example of initial and final configuration of such Turing machine if X= **aaababab** and Y=**abab.**is given in figure 3B

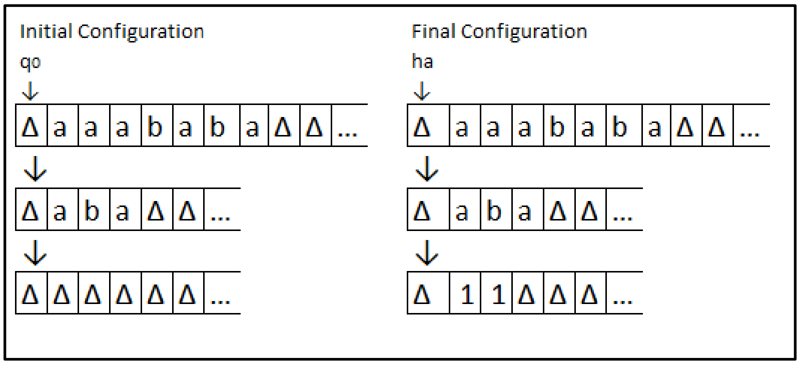


Figure 3B

NOTE: Output is 11 are there are two occurrences of Y in X (aaababab)

You also have to write a brief algorithm of your machine as well as create a state diagram.

Write your algorithm here:

Roll Number: ________________

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Theory of Automata	**Course Code:**	CS 301
	Program:	BS Computer Science	**Semester:**	Fall 2019
	Duration:	180 Minutes	**Total Marks:**	75
	Paper Date:	December 16, 2019.	**Weight**	45
	Section:	N/A	**Page(s):**	6
	Exam Type:	Final Exam		
Student : Name:______________________________		**Roll No.**________________		
Instruction/ Notes:	1. Solve in the space provided. Extra sheets will NOT be collected or marked. 2. One A4 handwritten help sheet is allowed. 3. In case of any ambiguity make a reasonable assumption. Good luck!			

Problem 1 **(Marks: 1+2+2)**

Given the regular expressions: $R_1 = (a^*+b)^*ab$ and $R_2 = (a+b^*)^*a^*b$

a. Write the three shortest strings generated by R_1.

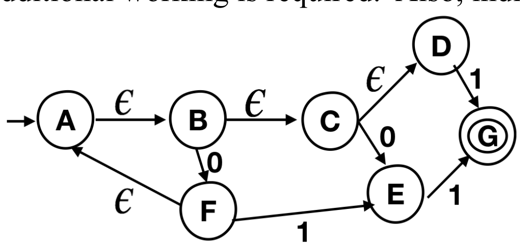
b. Give a regular expression for $L(R_1) \cap L(R_2)$

c. Which of the expressions are correct?

i. $L(R_1) \cup L(R_2) = \Sigma^*$ ii. $L(R_1) \subseteq L(R_2)$ iii. $L(R_1)L(R_2) = (a+b)^*ab(a+b)^*(a^*b^*)b$

Problem 2 **(Marks: 5)**

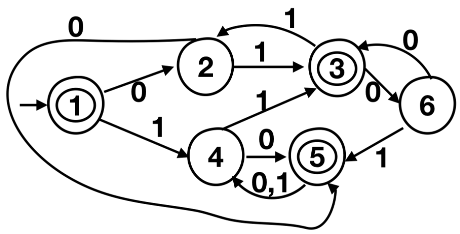
Remove all the ϵ-transitions from this NFA-ϵ and make the state transition diagram of the resulting NFA. No additional working is required. Also, indicate the **start and final states.**



State	0	1

Problem 3 **(Marks: 5)**

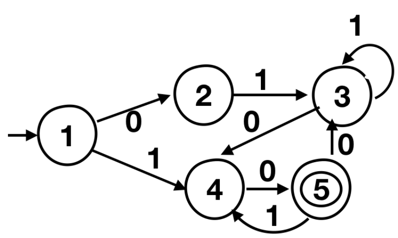
Minimize the following DFA. Fill out the table. Draw the resulting DFA and **clearly indicate which states were merged.**



6						x
5					x	x
4				x	x	x
3			x	x	x	x
2		x	x	x	x	x
1	x	x	x	x	x	x
	1	2	3	4	5	6

Problem 4 **(Marks: 5)**

Find the regular expression corresponding to the given automaton using the method of state elimination. Show all steps.



Problem 5 **(Marks: 5)**
L = $\{(0^n 1^{2n})* \mid n \geq 0\}$
Write the three shortest strings of L and express this language as a CFG. Clearly mark the start symbol.

Problem 6 **(Marks: 5)**
Use the CYK algorithm to determine if the string 0110 is generated by the given CFG (S is start symbol). Show working.
S → AZ | ZA
Z → 0 | AZ
A → 1 | ZA

Problem 7 **(Marks: 5)**
Prove that $n^4 2^{n^3} = 2^{O(n^3)}$. All steps are required.

Problem 8 **(Marks: 5)**

Prove that if we find a language A $\in$ P and we also find that A is NP-complete then P=NP

Problem 9 **(Marks: 5)**

Reduce the following instance of 3SAT to an instance of CLIQUE. Clearly specify the graph G and k.

CLIQUE = {<G,k> | G is an undirected graph with a k-clique}

$(x_1 \wedge x_1 \wedge x_2) \vee (\bar{x}_1 \wedge \bar{x}_2 \wedge \bar{x}_2)$

Problem 10 **(Marks: 5)**

Prove that the following language belongs to NP complexity class

L = {<G,s,t,k> | G is an undirected graph and there is a path from s to t of length at least k and no node repeats itself in the path}

Problem 11 **(Marks: 2x8=16)**

For each part, give a **one/two line explanation** or counter example. **Only yes/no answer scores zero.**

i. Can an NFA can be reduced to a DFA in polynomial time or not?

ii. If L_1 and L_2 are context free languages then is $L_1 \cap L_2$ also a context free language?

iii. Are all context free languages member of NP? Are they NP-complete?

iv. If $L_1 \subseteq L_2$ and L_1 is not a regular language then is L_2 also non-regular?

v. Are there any recursively enumerable languages member of NP?

vi. L = {<M,w> | M is a Turing machine and M halts on input w}. Is L Turing recognizable?

vii. Is $P \subseteq NP$?

viii. Can we simulate a PDA using a queue, which is a first in first out structure, instead of a stack?

Problem 12 **(Marks: 2+2)**

i. Write the complement of L defined over the alphabet $\{0,1\}$. L = $\{0^n1^n \mid n \geq 0\}$

ii. Suppose $L_1 = \{0^n1^n \mid n \geq 0\}$ and $L_2 = \{1^n2^n \mid n \geq 0\}$
Write down L_1L_2

Problem 13 **(Marks: 5)**

Make a deterministic single tape Turing machine to decide the following language:

L = {x | x ∈ {0,1}* and length of x is odd and there is a zero in the center of the string}

Examples of string in L are 11**0**10, 0111**0**0011, 0**0**1, etc.

Label each edge using Sipser's notation: Symbol read -> symbol written, {L,R} (you can omit symbol written if input symbol is not changed)

National University of Computer and Emerging Sciences, Lahore Campus

	Course: Program: Duration: Paper Date: Section: Exam:	Theory of Automata BS(Computer Science) 180 min Feb 13, 2021 ALL Final	Course Code: Semester: Total Marks: Weight: Page(s): Roll No.	CS301 Fall 2020 70 45% 10

Instructions/Notes: Please show all the working/reasoning, no marks will be given without it. Please attempt questions on the question paper, in the given space.
You can use rough sheets but those will not be collected.

Question 1 (10 points): Decidability

Identify if the following decision problem is decidable or not. Justify your answer by first providing a high-level algorithm (or TM) for this problem and then specify for what type of inputs the algorithm will Accept/Reject/Loop Forever.

Given two Deterministic Finite Automaton, M_1 and M_2, identify if $L(M_1) \cap L(M_2)$ is empty or not?

Note:
L(M) means the language of M, that is, a set of strings accepted by M.
$\cap$ is a symbol of *set intersection*.

Solution:
Algorithm: Input M1 and M2
Steps:

1. Create a DFA M of $L(M_1)$ and $L(M_2)$ using the closure properties of intersection of FAs. 2. Find a path from initial state to final states of M. (Path can be found using BFS).

3. If there is a path form initial state to final state/s Accept else Reject.

The problem is decidable as the algorithm will reject or accept for every input, and will not loop forever for any input.

Name:—————— Roll Number:——————

Question 2 (15 points): Designing Turing machine

Assume Σ = {0, 1}. Design a Single Tape Deterministic Turing Machine that sorts the input tape such that all 0's (if any) must appear before all 1's (if any).

For example, if the input tape is initially.

↓

Δ	0	1	1	0	0	1	1	Δ

Δ ...

Then once your Turing Machine halts, the tape should be

Δ	0	0	0	1	1	1	1	Δ

Δ ...

Where Δ = = Blank

Briefly describe steps of your algorithm in the space given below. Do not give a 1 or 2 line statement. Write a proper algorithm.

Then draw a state diagram of TM in space provided on the next page. Your TM design must have a maximum of 6 states, including ACCEPT (h_a) state.
Assume that initially pointer head (tape head) is placed on the left-most tape square that contains a Δ, followed by input string (as shown in the example initial input tape above). Tape head can be at

any position when the machine stops.

Solution:

Option 1:

Swap consecutive 10 in input iteratively until input is sorted.

Steps:

1. On state q0, keep moving right if 0 found on the input tape.
   a. if blank found , ACCEPT
2. Once a 1 is found on the input tape, move to state q1 and keep moving right on your input tape. a. If blank found, ACCEPT
3. If a 0 is found, convert it into 1 and move left.
4. Convert the 1 into 0 and move left.
5. Keep moving left until blank is encountered
6. Goto step 1

Option 2: Swap left most 1 with left most 0 on RHS of that 1 (if any).

Steps:

Start:
Find the left most 1 and mark it as 1^*(if not found goto ha)
Find the left most 0 on RHS of 1^*
If found
Change 0 to 1
Go back and change $1^*$to 0
Goto Start
else
Change $1^*$to 1
Goto ha.

Name:—————— Roll Number:——————

Question 3 (10 points): Dry run Turing Machine

Consider a multi-tape (3 tape) Turing machine shown in Figure 1.

0, Δ, Δ --> 0, Δ, 0, R,S, R
1, Δ, Δ --> 1, Δ , 0, R,S, R 0, 0 , Δ --> 0, 0, 0, R,
R, R 1, 1 , Δ --> 1, 1, 0, R, R, R 0, 1 , Δ --> 0, 1,
0, R, R, R 1, 0 , Δ --> 1, 0, 0, R, R, R
1, Δ, 0 -->1, Δ, 1, L, S, L
0, Δ, 0 -->0, Δ, 0, L, S, L
0, 0, 0 -->0, 0, 0, L, S, L
1, 1, 0 -->1, 1, 0, L, S, L
1, 0, 0 -->1, 0, 1, L, S, L

q1

halt q2

1, 0, 0 --> 1, 0, 0, L, L, L
0, 0, 0 --> 0, 0, 1, L, L, L
1, 1, 0 --> 1, 1, 1, L, L, L
0, Δ, 0 --> 0, Δ, 1, L, S, L

Δ, Δ, Δ, --> Δ, Δ, Δ, L, L, L Δ, Δ, Δ, --> Δ, Δ, Δ, R,R, R 1, Δ, 0 , --> 1, Δ, 0, L, S, L

1, 0, 0, --> 1, 0, 0, L, L, L q3

q

a, b,c, --> x, y, z, R, L, S

q p

a b c --> x y z , R L S

q p a --> x, R | b --> y, L | c --> z, S

a, b, c, / x, y, z, R, L, S

q p

0 , 1 , 0 , --> 0 , 1, 1 , L, L, L

q p

(a) Turing Maching for Question 3.

(b) Different notations for writing transition.[1]

Figure 1: Question 3, [1]*Note that you might have used a different way to write transitions of TM in class. For your ease please see notations given in (b), they all mean the same. Where a,b,c are read and x,y,z are written on tape 1,2,3 resp. And R,L,S are tape 1,2,3 heads' movement resp.*

When machine starts, the contents and the tape heads are shown below.

	↓										
Δ	1	1	0	0	1	0	1	0	0	Δ	...

	↓										
Δ	1	1	0	0	1	Δ	Δ	Δ	Δ	Δ	...

	↓										
Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	...

Where Δ = =Blank

a). Run the Turing machine on it and show your working and the final configuration of tapes. Write your final answer in empty tapes provided below. Your answer should show the final position of tape heads and tapes contents. Show the working on space provided on next page (no marks without showing working).
SOLUTION:

	↓										
Δ	1	1	0	0	1	0	1	0	0	Δ	...

↓

Δ	1	1	0	0	1	Δ	Δ	Δ	Δ	Δ	...

Δ	1	0	1	1	1	1	0	1	1	Δ	...

b). Write down, in the box below, what calculation is this Turing Machine (Figure 1a) performing? Your answer should be of one line only.

SUBTRACTION

Space to show working of running machine

Question 4 (10 points): NFA to DFA

Convert the NFA given in Figure 2 to DFA. Draw the Transition table as well as the State diagram of the resulting DFA. The set of alphabet, Σ = {0, 1}

Note: Your final DFA must not contain any unreachable states.

q0 1 0

q1

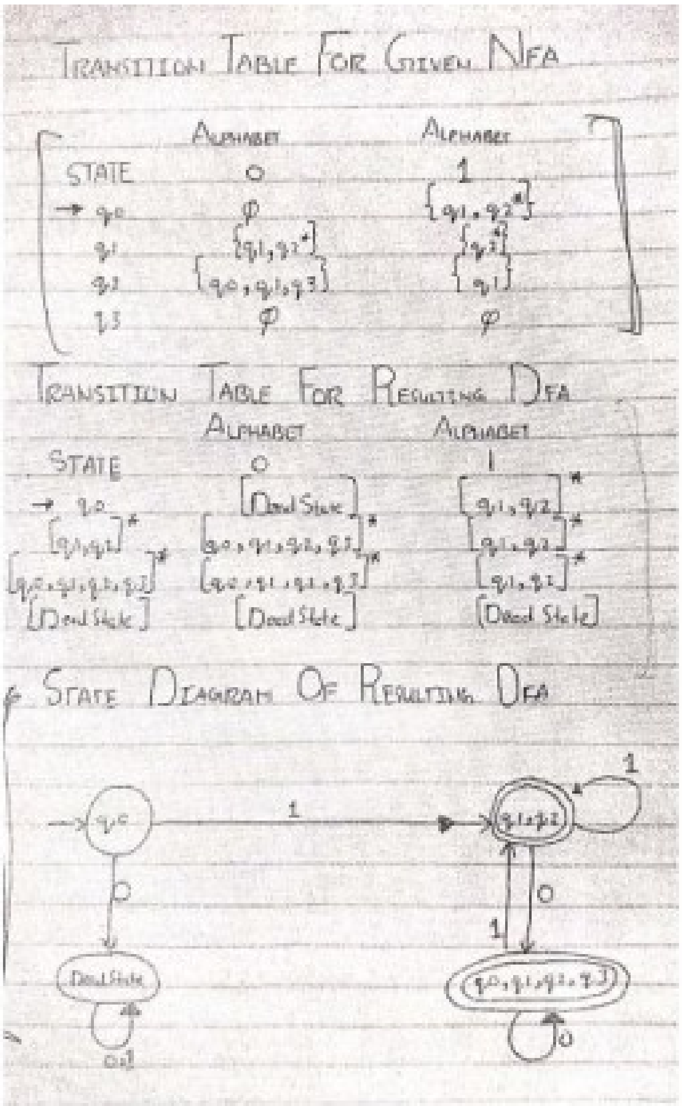
1

0

q3

q2 0

Figure 2: NFA for Question 4



Transition Table For Given NFA

STATE	Alphabet 0	Alphabet 1
→ q0	φ	{q1, q2*}
q1	{q1, q2*}	{q2*}
q2	{q0, q1, q3}	{q1}
q3	φ	φ

Transition Table For Resulting DFA

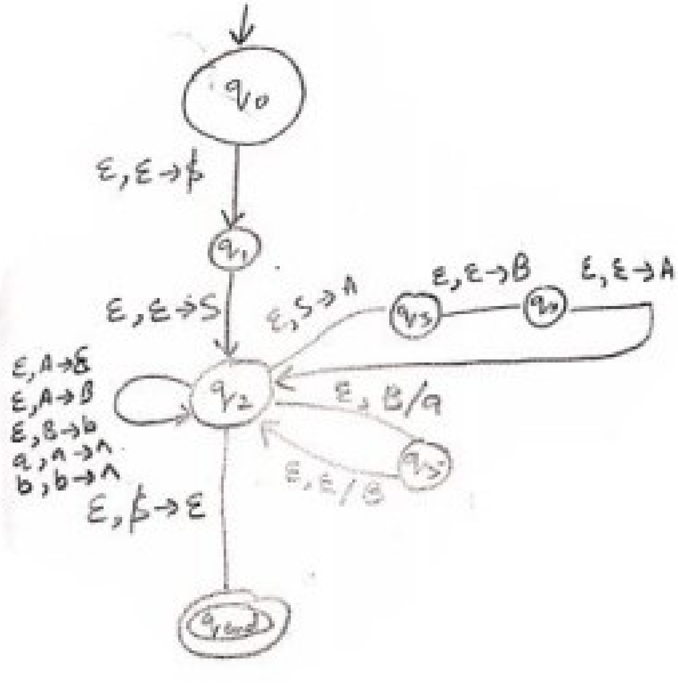
STATE	Alphabet 0	Alphabet 1
→ q0	[Dead State]	[q1, q2]*
[q1, q2]*	[q0, q1, q2, q3]*	[q1, q2]*
[q0, q1, q2, q3]*	[q0, q1, q2, q3]*	[q1, q2]*
[Dead State]	[Dead State]	[Dead State]

State Diagram Of Resulting DFA

Question 5 (5 points): CFG to PDA

Using the proof of the theorem *"Any language accepted by a CFG has an equivalent PDA"*, convert the following CFG to PDA.

S → ABA
A → ε | B
B → Ba | b
Note: $\varepsilon = \lambda$ = null



Question 6 (10 points): Designing CFG

Construct a CFG for the language L consisting of words w_i. The alphabet set is {a, b}. Every word w_i = xyz i.e. it can be split into substrings x, y, and z *where*

x contains even number of a's and any number of b's. $|x| \geq 0$. Some examples of x are *aa, abaaa, bbbaa*, etc. y consists of odd number of a's and any number of b's. y can also be ε. So, $|y| \geq 0$. Some examples of y are *ε, a, abaa, bbba*, etc.
z is of the form $a^m b^n$ where $m \leq n \leq 2m$ and $|z| \geq 0$.

While designing the CFG, you must also fulfill the following constraints.

1. At max, you can only use 3 non-terminals (also know as variables).
2. The CFG must begin with the non-terminal S.
3. The CFG has 3 lines.

4. At max, you can use 4 ‘|’ symbols in CFG.

Some valid words for this language are *aaaab, aaab, abaaabbaabb, abbbab*, etc.
Note: $\varepsilon = \lambda$ = null

Solution:

The idea is to consider x.y together as $(a+b)^*$

S → MN
M → aM | bM | ε
N → aNB | aNbb | ε

Question 7 (10 points): Pumping Lemma for CFL

Using Pumping Lemma for CFL, show that the following language is not a context free

language. L= { $a^i b^j c^k$| $k= i*j$ and $i, j, k \geq 0$}

Assume Σ={a, b, c}

3) vxy contain all cs $\Rightarrow$ if $i=2$ $uv^2xy^2z = a^p b^p c^{p^2+|v|+|y|}$

as $|vy| \geq 1$ $n(a) * n(b) \neq n(c)$

4) vxy contain some as and some bs :

↳ if v contains on as & y contains on bs

$i=2$ $a^{p+|v|} b^{p+|y|} c^{p^2}$ $n(a) * n(b) \neq n(c)$

↳ if v or y contain string ~~ab~~ both ab .
pumping up will change the order and
new strings with no be in L .

5) vxy contain some bs and some cs

↳ if v contains only b's and y contains only cs .
$i=2$ $a^p b^{p+|v|} c^{p^2+|y|}$

$n(a) * n(b) \neq n(c)$ because $|y| \leq p$

↳ if v or y contain some bs and some c
then pumping up will change the order & new string

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therefore L is not a [illegible]

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National University of Computer and Emerging Sciences, Lahore Campus

Course:	Theory of Automata	Course Code:	CS-3005
Program:	BS (Computer Science)	Semester:	Fall 2021
Duration:	180 minutes	Total Marks:	75
Paper Date:	21st Jan 2022	Page(s):	10
Section:	ALL	Section:	
Exam:	Final Exam	Roll No:	

Instruction/Notes: Answer in the space provided, showing all the working
NO ROUGH SHEETS
In case of any confusion or ambiguity make a reasonable assumption. Good luck!

Question 1: Prove that the language L is non-regular using the **pumping lemma for regular languages**. Your solutions should clearly show a string, length of string, all the possible divisions of the strings, and contradiction for all the divisions **(10 points)**

L = {www | w ∈{0,1}*}

Suppose L is RL with pumping length p.

$s = 0^p 1 0^p 1 0^p 1$ $|s| > p$.

possible divisions to uvw s.t. $v \neq \Lambda$ & $|uv| \leq p$

v will get one or more zeros from first 0^p.

$\underbrace{0^{p-n-m}}_{u} \underbrace{0^n}_{v} \underbrace{0^m 1 0^p 1 0^p 1}_{w}$.

where $n > 0$

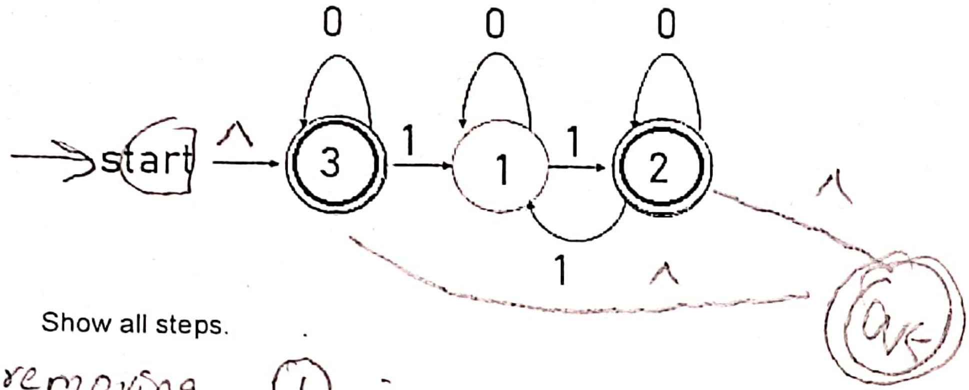
for $i = 0$ $uv^0w = 0^{p-n} 1 0^p 1 0^p 1 \notin L$

there fore $L \notin RL$.

→ marks deducted if value of p is assumed.
→ if division not done s.t. $|uv| \leq p$.

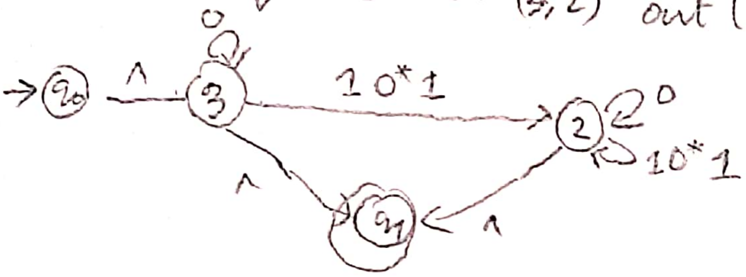
Roll Number:____________ Section: ____________

Question 2: Convert the following FA to RE using the **state elimination method**. Eliminate the states in the given order: 1, 2 and then 3. **(10 points)**

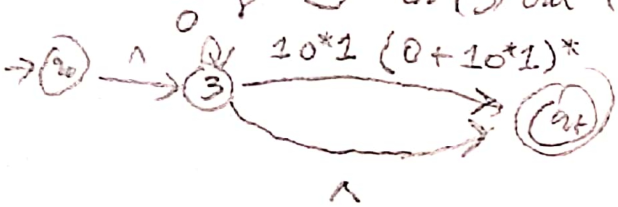


Show all steps.

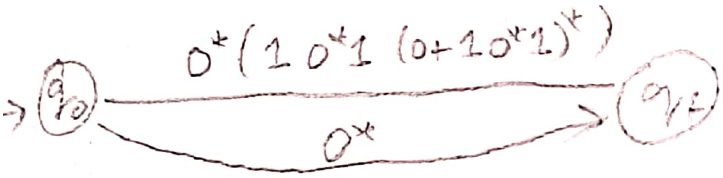
removing (1) in (3, 2) out (2)



removing (2) in (3) out (2)



removing (3) in q_0 out q_f



The RE given in this box will only be marked

$0^* + 0^*(10^*1(0+10^*1)^*)$.

$0^*(\lambda + 10^*1(10^*1+0)^*)$

Roll Number:______________ Section: __________

Question 3: You have a multi-tape Turing machine (MTM) that has 4 tapes T1, T2, T3, and T4. Initially, T1, T2 and T3 contain one word/string each, while T4 is empty. Design a MTM that concatenates the words *w1*, *w2* and *w3* on tapes T1, T2 and T3 respectively, and store the resulting word on T4 in the form *Δw1#w2#w3#*. Input alphabet set = Σ = {a,b}. All tapes heads are initially at the leftmost slot. Consider the sample example:

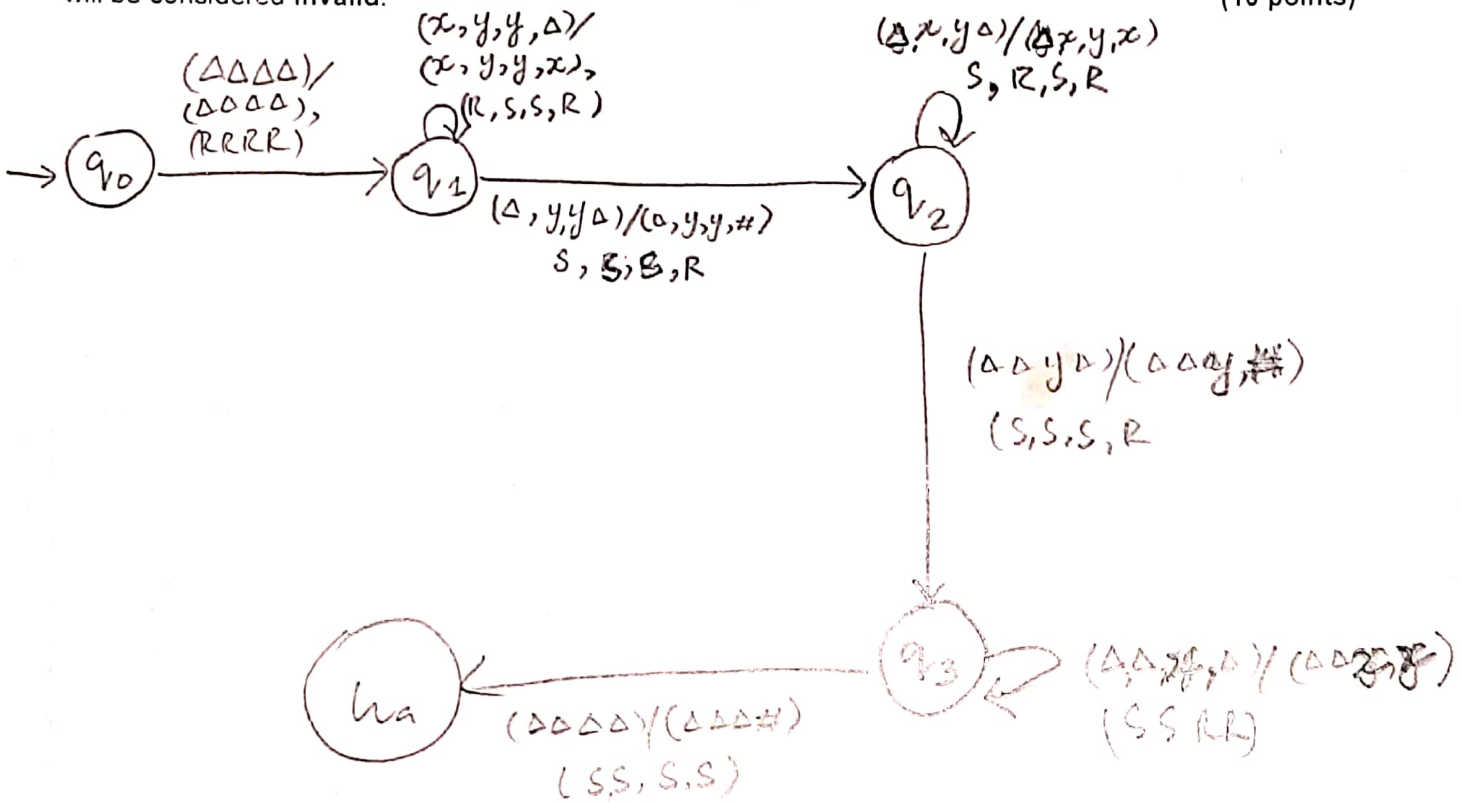
Input:

T1	Δ	a	b	Δ		
T2	Δ	b	b	a	Δ	
T3	Δ	b	Δ			
T4	Δ	Δ	Δ	Δ	Δ	Δ

Output:

T4	Δ	a	b	#	b	b	a	#	b	#

You must use as few states as possible. An MTM with more than 5 states **including** the start and final states will be considered **invalid**. **(10 points)**



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Roll Number:____________ Section: ____________

Question 4: Convert the following CFG to CNF. Neatly write the final CFG in the table below. (10 points)

S-> ABA | C | Cb
A-> λ
B->ab
C->B | λ
D->a
Note: λ **represents null**
Show all steps:

Step 1:- Remove $A \to \lambda$

$S \to B \mid C \mid Cb$ (not adding AB, BA as A only goes to Null)

A

$B \to ab$

$C \to B \mid \lambda$

$D \to a$

Remove $C \to \lambda$.

$S \to B \mid C \mid \lambda \mid Cb \mid b$

$B \to ab$

$C \to B$

$D \to a$

Step 2 Remove unit Productions -

$S \to ab \mid \lambda \mid Cb \mid b$.

~~B → ab~~ [use less now]

$C \to ab$

$D \to a$

Step 3:- Remove. two terminals & terminal & variable

$S \to X_a X_b \mid \lambda \mid CX_b \mid b$

$C \to X_a X_b$.

$X_a \to a$ $X_b \to b$.

The Final CFG given in this box will only be marked

$S \to X_a X_b \mid CX_b \mid b \mid \lambda$

$C \to X_a X_b$

$X_a \to a$

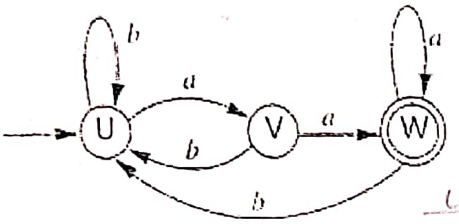
$X_b \to b$.

Roll Number:______________ Section: __________

Question 5: Given D1 and D2, determine DFA of L(D1) - L(D2)' using the proofs of closure properties. For credit, provide the final DFA in the box given on the next page. **(10 points)**

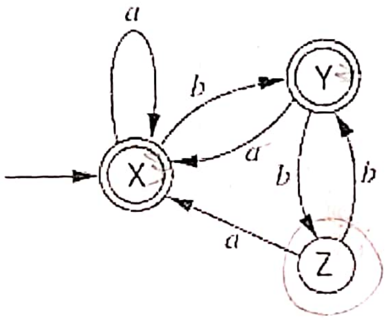
Note: ' is complement

D1:



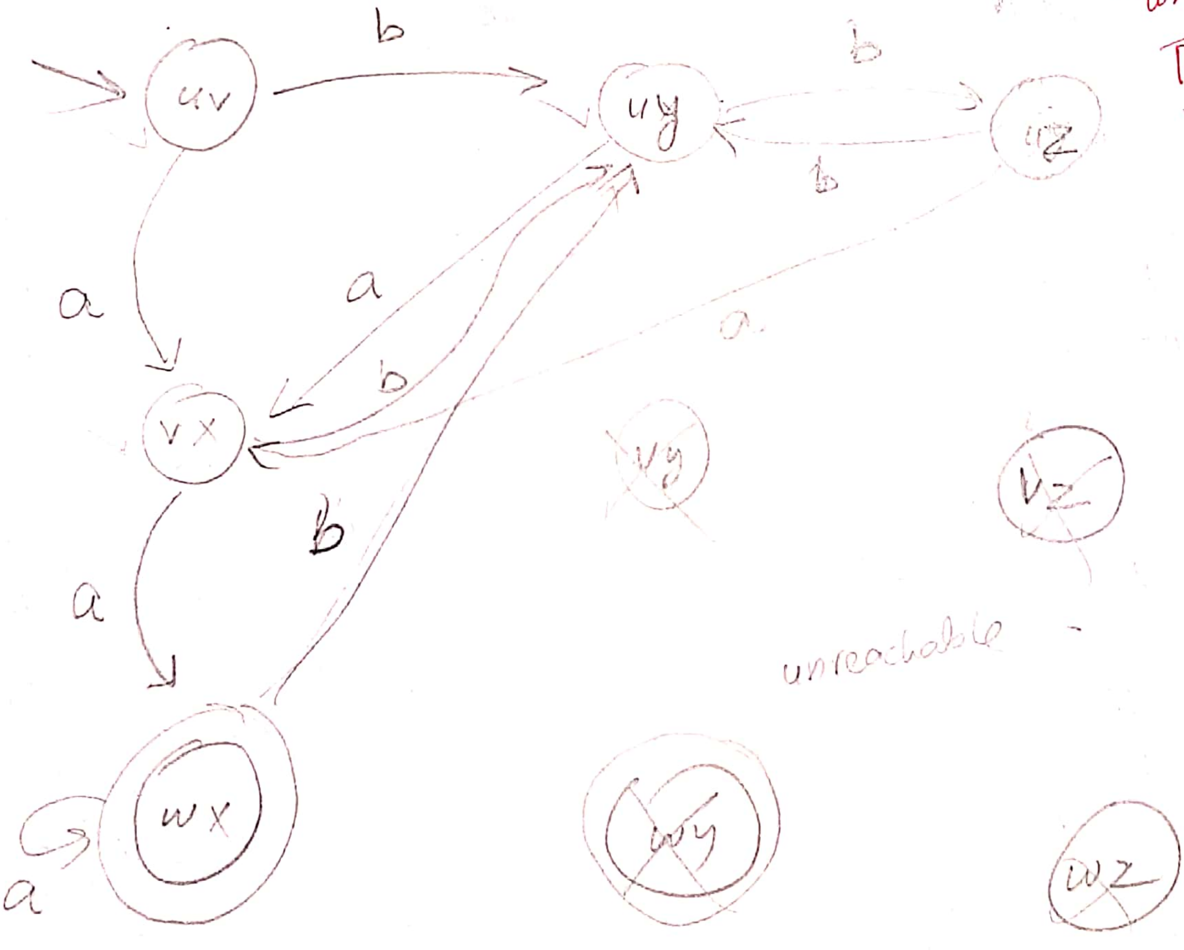
	a	b
U	V	U
V	W	U
W	W	U

D2: D2'



	a	b
x	x	y
y	x	z
z	x	y

Show all steps.



has to be done using closure properties. otherwise 0 marks.

D_2' 2 points. rest is for $D_1 - D_2'$

final states are final of D1 {w}

& non final of D_2' {x, y}

Question 6: Short questions:

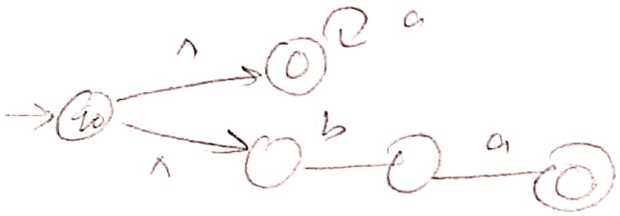
a: What is the use of Pumping Lemma for CFL, write a 1-line answer **(3 points)**

To ~~show~~ prove that a language is non CFL

b: Convert the following RE to NFA-null. Show all steps **(5 points)**

a*+ba

a* + ba



c: Given a CFG write in one line what is the language of this CFG. **(5 points)**

Note: λ represents null.

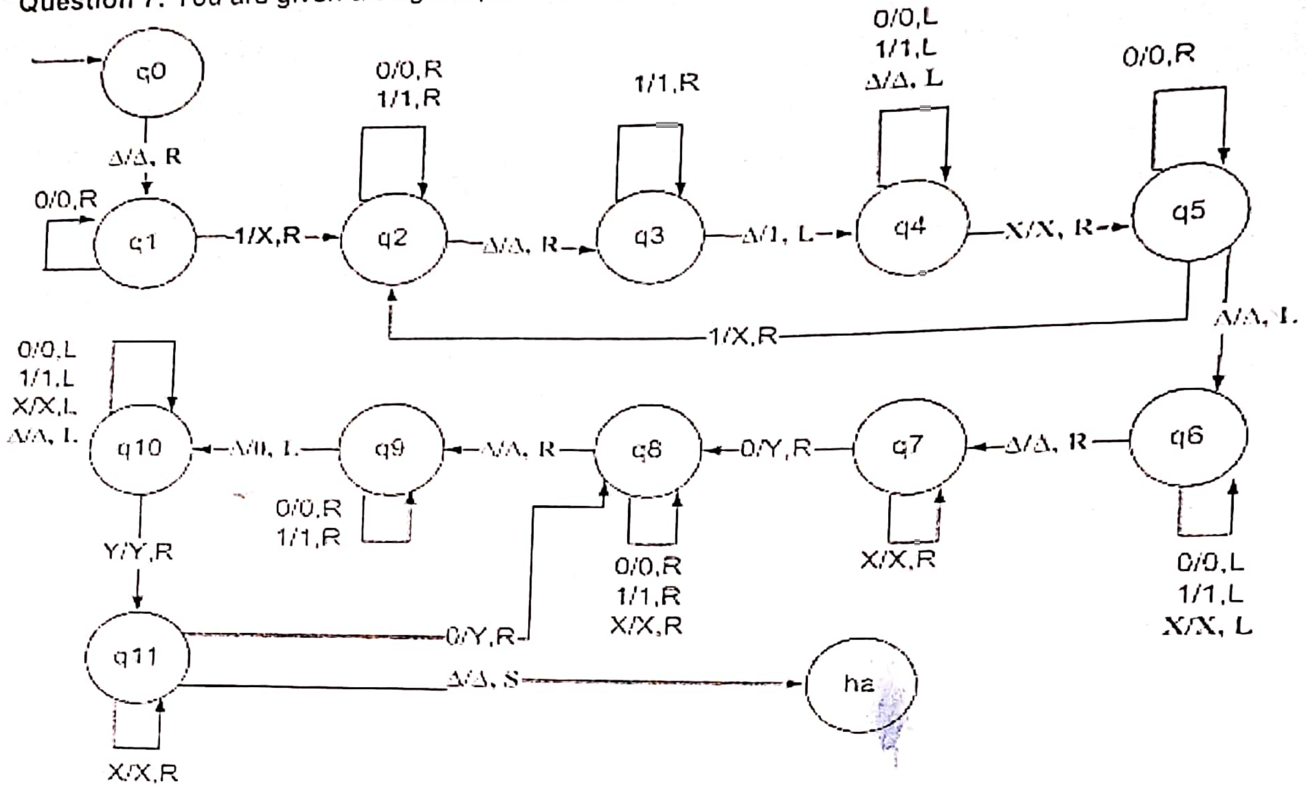
S → Aa | MS | SMA
A → Aa | λ
M → λ | MM | bMa | aMb

Strings with $n_a > n_b$

Section: ____________

Roll Number:____________

Question 7: You are given a single tape TM. The state q0 is the start state.



You have to run the TM on the string 11010101 and determine the configuration of the resulting tape? (Show working) **(12 points)**

Tape you are given:

Δ	1	1	0	1	0	1	0	1	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	...

Tapes for rough work:

Δ																				
Δ																				
Δ																				
Δ																				
Δ																				

Resulting tape:

Δ	X	X	Y	X	Y	X	Y	X	Δ	1	1	1	1	1	0	0	0	Δ	Δ	Δ

Description of TM: Briefly explain in 1 sentence what is the TM doing (answer should be generic).

Writes sorted string after input in desc order

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National University of Computer and Emerging Sciences, Lahore Campus

	Course:	Theory of Automata	Course Code:	CS-3005
	Program:	BS (Computer Science)	Semester:	Fall 2022
	Duration:	180 Minutes	Total Marks:	80
	Paper Date:	17-December-2022	Weight	40 %
	Section:	ALL	Page(s):	16
	Exam:	Final Term	Roll No.	

Instruction/Notes:
1. Answer in the space provided, showing all the work.
2. Rough Sheets are not allowed.
3. In case of confusion or ambiguity make a reasonable assumption.
4. Attempt all Questions

Section 1: (Short Question Answers) [25 Marks]

Q1: What is the cardinality of L? [3 Marks]

$L = \{ w \text{ over } \Sigma \mid |w| > 5 \text{ and } |w| \leq 10 \}$.

$\Sigma = \{0,1,2\}$

Note: Cardinality means the total number of elements in the given set.

88209

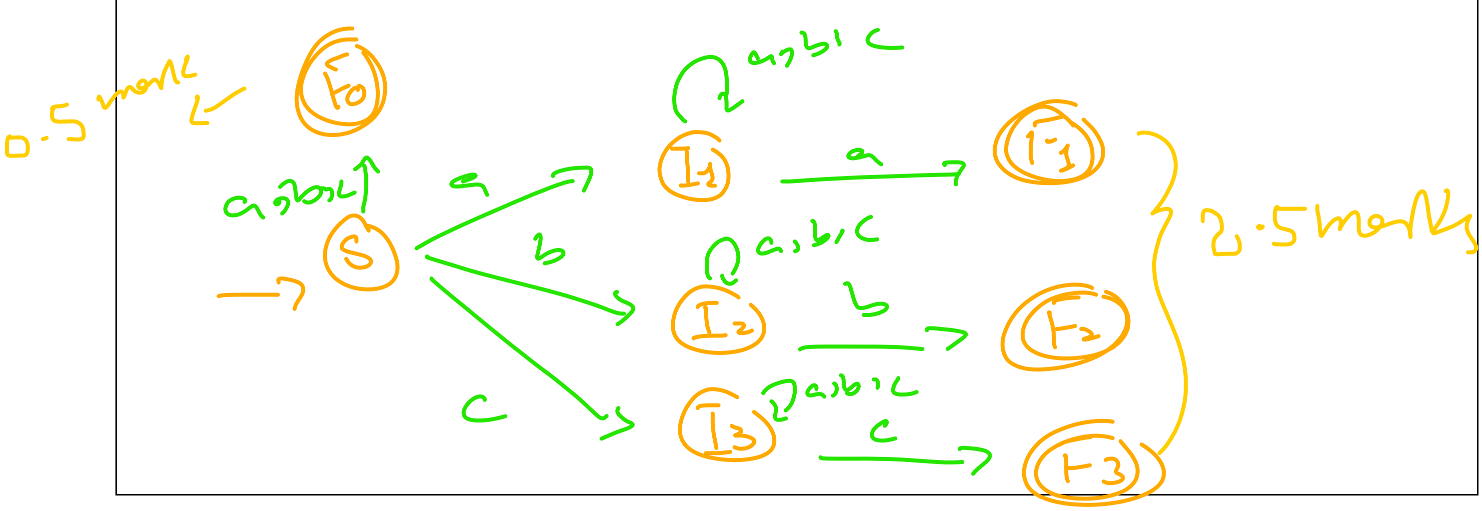
3 marks

$\Sigma^6 + \Sigma^7 + \Sigma^8 + \Sigma^9 + \Sigma^{10} = 3$

Q2: Design a Finite Automaton (DFA or NFA) for the following language. [3 Marks]

$L = \{ x \mid x \text{ over } \{a, b, c\}\ x \text{ starts and ends with same alphabet} \}$

Note: Pick a suitable sub-category from FA and design the machine accordingly.



Q3: What will be the language of the following grammar? [7 Marks]

$$L \rightarrow ALB \mid AABB$$

$$A \rightarrow aAb \mid aaabbb$$

$$B \rightarrow ccBd \mid \wedge$$

Note: You are required to write answer in a proper format. For Example, see Q1 statement. You are expected to write a proper answer based on CFG given above. Lengthy Statements are not required here.

3

$L = \{ (a^i b^i)^k (c^{2j} d^j)^k \; ; \; i \geq 3 ; j \geq 0 ; k \geq 2 \}$

2 2

Q4: Write a Regular Expression for the following Language. [4 Marks]

$$L = \{ x \mid x \text{ over } \{a, b\} \; x \text{ contains 'aba' and 'bab' as a substring} \}$$

R.E = { abab + baba + (a+b)*aba(a+b)*bab(a+b)* + (a+b)*bab(a+b)*aba(a+b)* }

1 1 1 2

Q5: Design the transition diagram of a PDA for the following Language? [4 Marks]

$$L = \{a^n b^m \; ; \; n + m = \text{even} \}$$

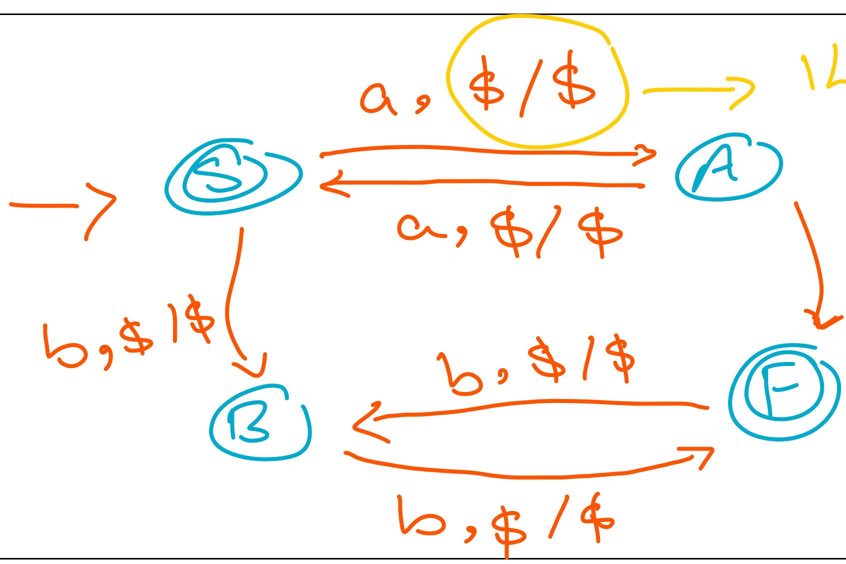
a's 1

b's 1

ab

↳ E+E 1

O+O 1



Key point without it you will get '0' marks

Q6: What will be the Regular Expression for the following Finite Automaton? [4 Marks]

Start State = A & Final State = {A,C}

States(q)	δ(q,a)	δ(q,b)
A	B	A
B	B	C
C	D	C
D	D	D

Note: Use State Elimination Method for extraction of Regular Expression. Write Final Regular Expression in the space provided below. Delete the given states in the following order, first State A then B then C & then D.

Regular Expression:

$$b^* + \overbrace{b^* a^+ b^+}^{3} \;/\; b^*(\overset{1}{\lambda} + a^+ b^+)$$

1

Exact marks (4)

3

No partial marks

Section 2: (Long / Detailed Solving Question Answers) [55 Marks]

Q1: Develop 3 multi-tape TM having 2 inputs X and Y (X and Y ε {0,1}*) [15 Marks]

X is on tape 1 and Y is on tape 2. Y slides over the X with the step of 1. Each time it computes the exclusive nor (XNOR) of the corresponding overlapping bits and note down the number of 1's (only) in tape 3 as shown below:

A	B	A XNOR B
0	0	1
0	1	0
1	0	0
1	1	1

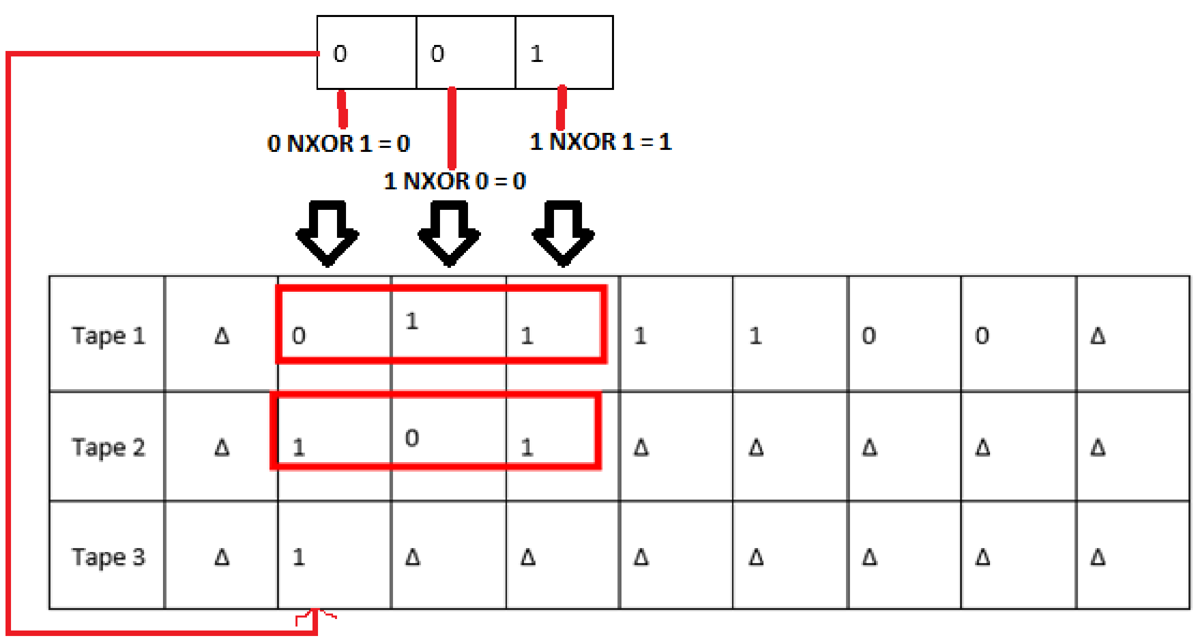
Truth Table for XNOR for inputs A and B

Initial configuration of 3 multi-tape TM

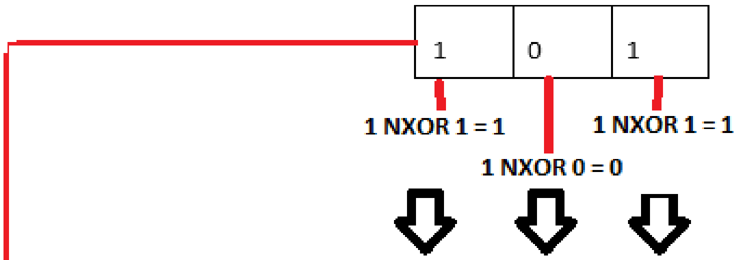
Tape 1: X	Δ	0	1	1	1	1	0	0	Δ
Tape 2: Y	Δ	1	0	1	Δ	Δ	Δ	Δ	Δ
Tape 3: Output	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ

Y will slide 5 times on X (in this example)

First time (first slide)



Second time (second slide)



Tape 1	Δ	0	1	1	1	1	0	0	Δ
Tape 2	Δ	1	0	1	Δ	Δ	Δ	Δ	Δ
Tape 3	Δ	1	1	1	Δ	Δ	Δ	Δ	Δ

.

.

.

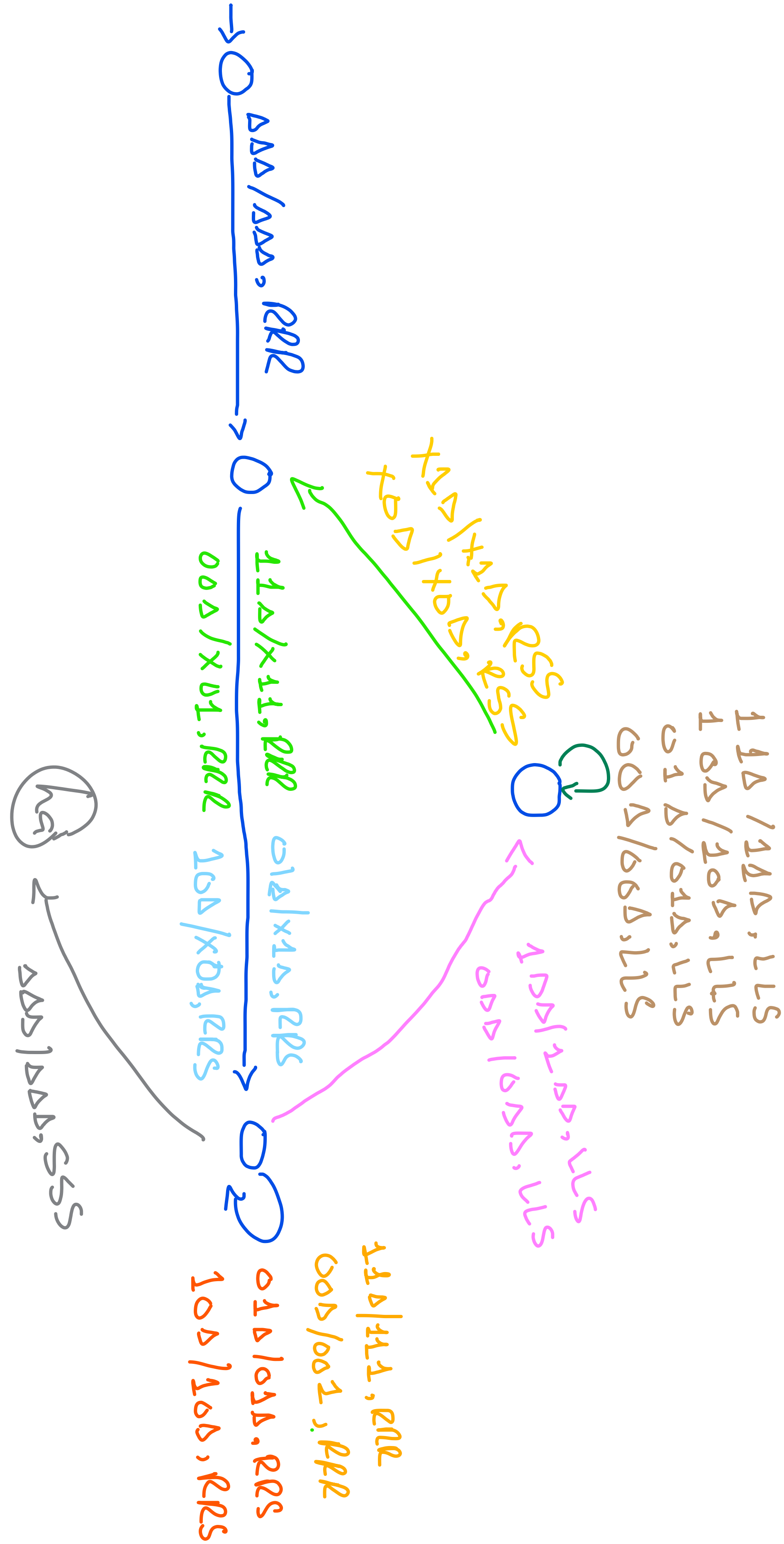
(last and 5th slide) Eventually Output will be

Tape 1	Δ	0	1	1	1	1	0	0	Δ	Δ
Tape 2	Δ	1	0	1	Δ	Δ	Δ	Δ	Δ	Δ
Tape 3	Δ	1	1	1	1	1	1	1	1	Δ

Provide the algorithm first that will explain your logic in simple statement and then draw TM:

Note: ***Be clear and to the point***. Clearly mention where your pointers are. **No marks if algorithm is incorrect.**

Algorithm:


▷▷▷/▷▷▷, RRR
11▷/X11, RRR
00▷/X01, RRR
01▷/X1▷, RRS
10▷/X0▷, RRS
11▷/111, RRR
00▷/001, RRR
01▷/011, RRS
10▷/100, RRS
1▷▷/1▷▷, LLS
0▷▷/0▷▷, LLS
11▷/11▷, LLS
10▷/10▷, LLS
01▷/01▷, LLS
00▷/00▷, LLS
X1▷/X1▷, RSS
X0▷/X0▷, RSS
▷▷▷/▷▷▷, SSS

Q2: Dry run the single-tape Turing machine on page 10 and give the content of the tape after running it (When TM halts). [15 Marks = 10 + 5] ?

The initial configuration of the TM is given below

Δ	1	0	1	1	Δ	0	0	1	0	Δ	Δ	.....	Δ

⇧ head/pointer

1 pointer

Answer:

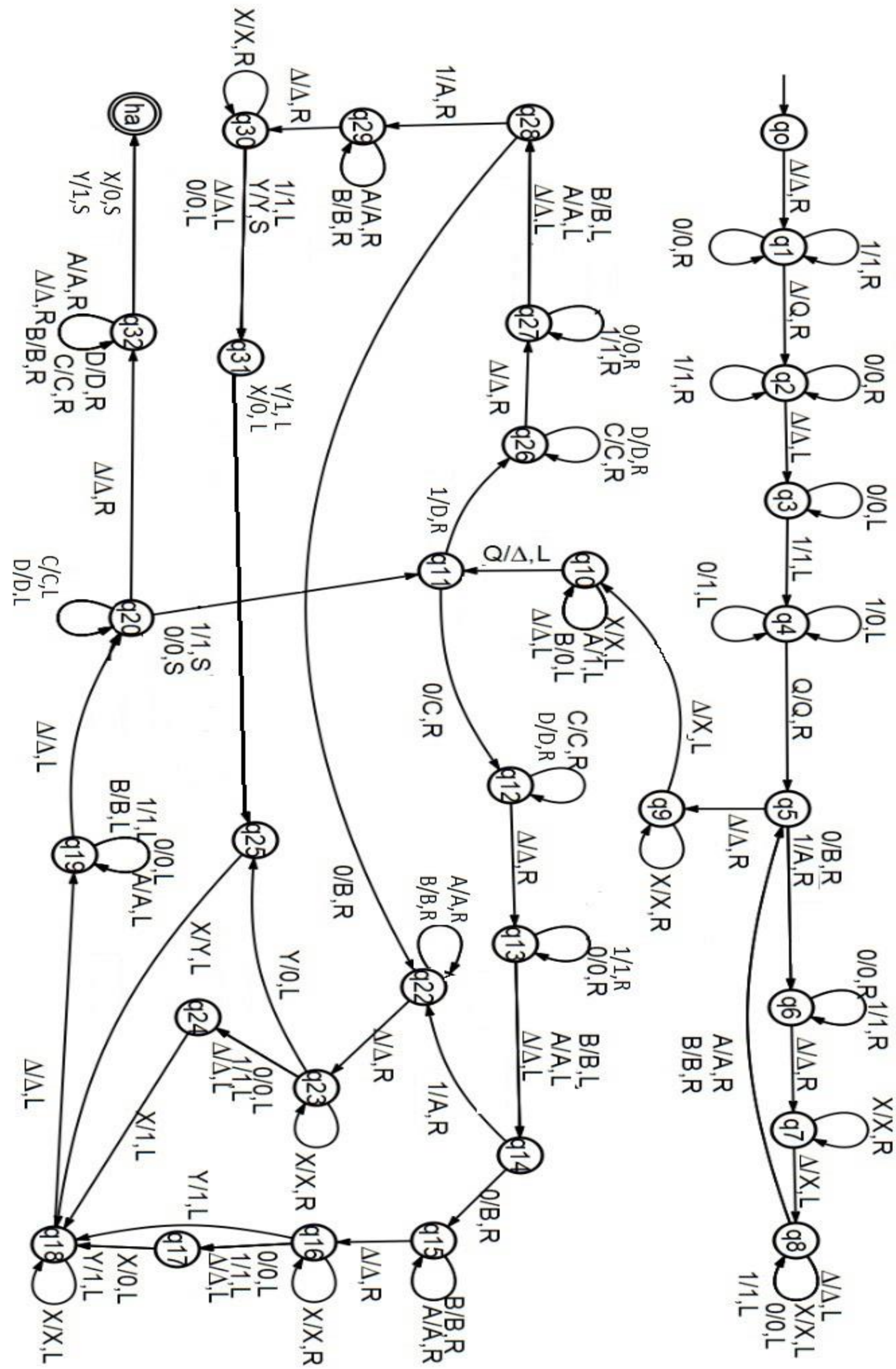
2

Δ	D	C	D	D	Δ	A	A	A	B	Δ	1	1	0	0	1	Δ			

Clearly show where will be the head/pointer when TM halts

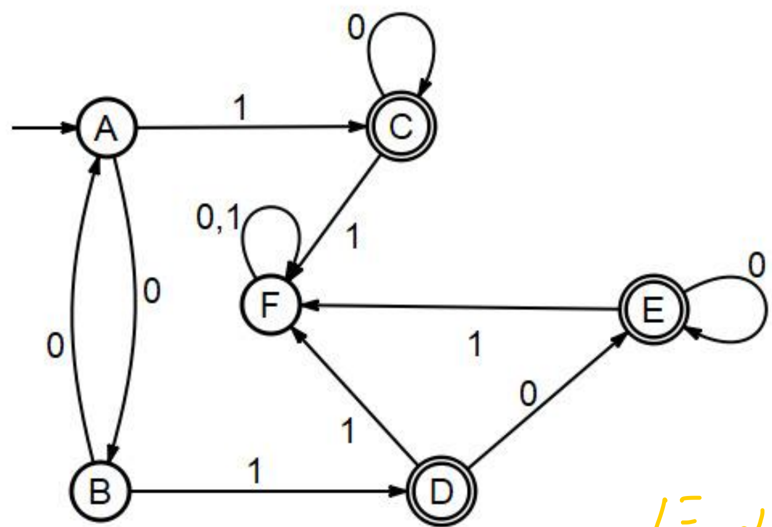
2 5

What is TM doing? (Explain in not more than 2 lines. Be brief and to the point. No mark for stories)


q0
q1
q2
q3
q4
q5
q6
q7
q8
q9
q10
q11
q12
q13
q14
q15
q16
q17
q18
q19
q20
q22
q23
q24
q25
q26
q27
q28
q29
q30
q31
q32
ha

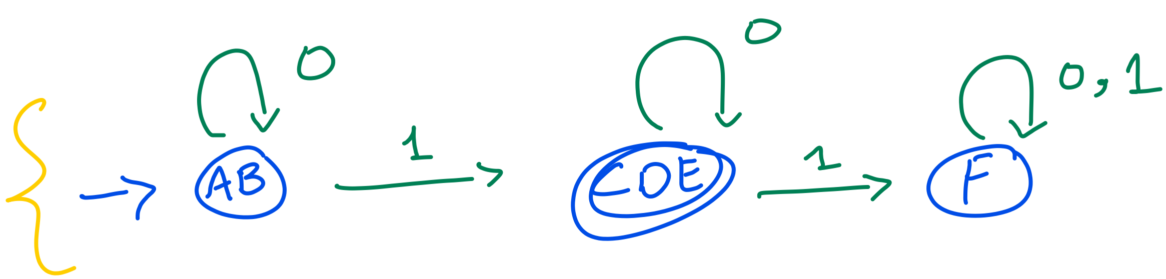
Q3. For the DFA pictured in the figure below, use the minimization algorithm discussed in the class to find a minimum-state DFA recognizing the same language. [10 Marks]

DFA:



Each raw = 1 mark

Minimized DFA:



4 marks

R.E = $\{0^* 1 0^*\}$

	A	B	C	D	E	F
A	-					
B		-				
C	✓	✓	-			
D	✓	✓		-		
E	✓	✓			-	
F	✓	✓	✓	✓	✓	-

A,B

C,D

C,E
D,E

Possible Grouping

1x6 = 6

<u>Note:</u> Use only the cell required

Q4. Let G be the following CFG: 3 marks

$S \rightarrow AaB \mid aB$
$A \rightarrow a \mid Aa$
$B \rightarrow b \mid C$
$C \rightarrow bC \mid a$

X -> a
Y -> b
Z -> AX
S -> ZB | XB
A -> AX | a
B -> YC | a | b
C -> YC | a

CNF Grammar

Determine whether the string "abba" is a member of L(G) using CYK Algorithm. [10 Marks]

j=4	S	-	-	-
j=3	-	B,C	-	-
j=2	S	-	B,C	-
j=1	X,A,B,C	Y,B	Y,B	X,A,B,C
	a	b	b	a

Note: Use only the cell required

Each row = 1.5 marks = 1.5x4 = 6

'abba' belongs to Language. => 1 marks

Require Work for CFG (if needed)

Q5. Tell whether the following Language is context free (CFL) or non- context free (non- CFL). If it is CFL provide PDA else prove it using Pumping Lemma

$$L = \{a^{m^2} b^m \mid m \geq 0\}$$ [5 Marks]

$x = a^{p^2} b^p$

			$a^{p^2-r-s-t}$
$u =$	Λ	a^{p^2}	
$v^i =$	a^{p^2-r-s}	b^{p-r-s}	a^r
$w =$	a^r	b^r	a^s
$y^i =$	a^s	b^s	a^t
$z =$	b^p	Λ	b^p

. . . .

Selecting string = $\underline{1}$ marks

divisions = $\underline{2}$ marks

proof = $\underline{2}$ marks

Rough Work

National University of Computer and Emerging Sciences, Lahore Campus

Course:	**Theory of Automata**	**Course Code:**	**CS-3005**
Program:	**BS (Computer Science)**	**Semester:**	**Fall 2023**
Duration:	**180 Minutes**	**Total Marks:**	**60**
Paper Date:	**27-December-2023**	**Weight**	**40 %**
Section:	**ALL**	**Page(s):**	**12**
Exam:	**Final Term**	**Roll No.**	

Instruction/Notes:
1. Answer in the space provided, showing all the steps.
2. You can take rough Sheets but will not be collected.
3. In case of confusion or ambiguity make a reasonable assumption.
4. Attempt all Questions

CLO 1				CLO 2			CLO 3			CLO 4		
a	b	c	**Total**	a	b	**Total**	a	b	**Total**	a	b	**Total**
3	1	8	**12**	2	10	**12**	4	8	**12**	10	14	**24**

CLO 1 [3 + 1 + 8 =12 Marks]

Question 1:

a) If L_1, L_2 and L_3 are Context free languages and $L_4 = L_1 \cap (L_2 \cup L_3)$. What kind of language will be L_4. (RL, CFL or non-CFL) Explain briefly [3 Marks]

Non CFL
$L_2 \cup L_3$= CFL
$L_1 \cap (L_2 \cup L_3)$= Non CFL

b) **True/ False** Context free languages are closed under difference. [1 Marks]

False

c) Tell whether the following Language is context free (CFL) or non- context free (non- CFL). If it is CFL provide PDA else prove it using Pumping Lemma.[2 + 6 = 8 Marks]

$$L = \{a^i\, b^j\, c^i\, d^{(i+j)} : i, j \geq 0\}$$

same approach we discussed during the class.

CLO 2 [2 + 10 =12 Marks]

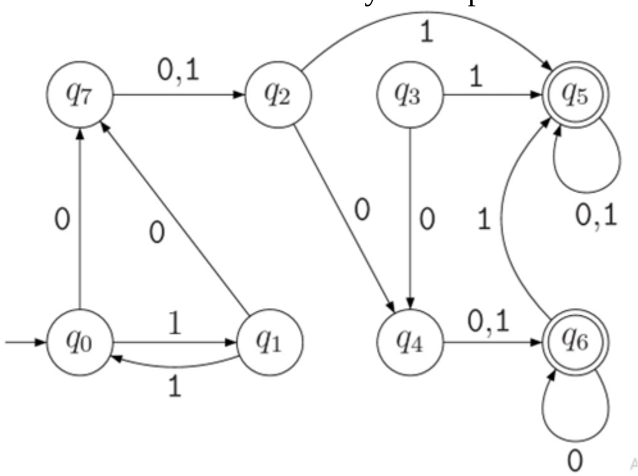
Question 2:

a) **Tick all regular expressions which express** [2 Marks]

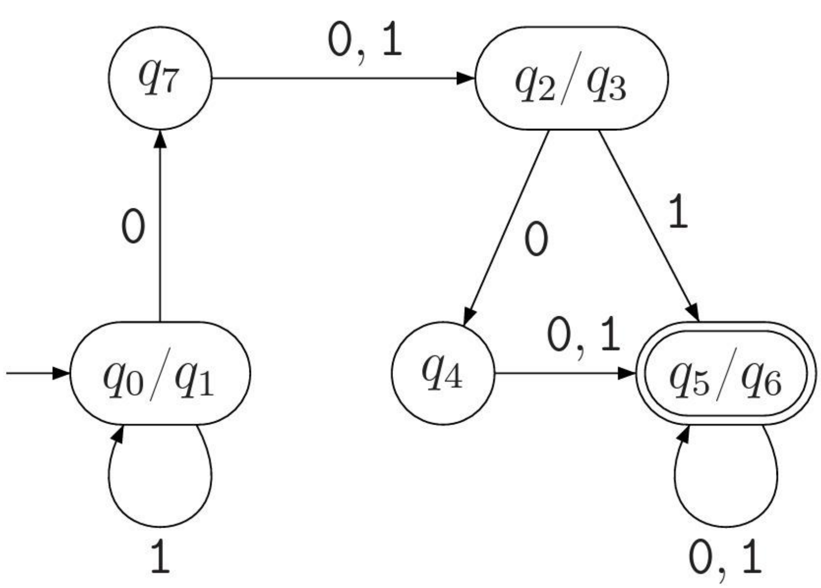
L = {w | w ∈ {0,1}* and w has no consecutive 0 and no consecutive 1}

a. (10)*+(01)*
b. (10 + 01)*
c. (0(10)* + 1(01)*)*
d. None of these

b) **Minimization of DFA** [5 Marks]
Find a minimum-state DFA recognizing the same language. Show complete working. Use only the method discussed in your respective classes.



Minimal DFA



CLO 3 [4 + 8 =12 Marks]
Question 3:

a) Tell whether the following grammar is ambiguous. How? Give justification [2+2 = 4 Marks]
$S \rightarrow 0S1 \mid 0S11 \mid \Lambda$

Ambiguous
Justification: two LMD OR 2 RMD from same string WITH EXAMPLE

b) The Html Table Creator [6 + 1.5 = 7.5 Marks]

<table>
<tr><td>Statement:</td><td>Sample HTML Code</td></tr>
<tr><td>Consider a context-free grammar that generates Strings in a specific format to create an HTML table. The grammar produces a table with rows%3=0 contains Numbers only, where all other rows can contain a Number or Alphabet. The goal is to design a grammar that generates valid HTML code for such a table structure.

Sample Output:
<table>
<tr><td>2</td><td>4</td></tr>
<tr><td>A</td><td>5</td></tr>
<tr><td>1</td><td>D</td></tr>
<tr><td>8</td><td>9</td></tr>
</table>
</td><td><table>

<tr>
<td>2<\td>
<td>4<\td>
<\tr>
<tr>
<td>A<\td>
<td>5<\td>
<\tr>
<tr>
<td>1<\td>
<td>D<\td>
<\tr>
<tr>
<td>8<\td>
<td>9<\td>
<\tr>
<\table></td></tr>
</table>

Design a context-free grammar that generates HTML code for a table with rows%3=0. The multiple of 3 rows should display numbers in each cell, while all the other rows should display alphabet or number. Each row should have at least one cells/columns. Table must contain at least one row. So the minimum number of rows and columns could be a single cell that is one row and one column. Your CFG can have variable number of cells/columns for each row. The generated HTML code should follow the standard syntax and structure of an HTML table. For the context-free grammar, you need to define the production rules that generate the HTML code for the desired table structure. Consider the use of non-terminal symbols for different components of the HTML code, such as the <table>, <tr>, <th>, and <td> tags, as well as the number and alphabet elements. You can use this scenario to design and explore the grammar rules that would generate the desired HTML table structure.

Solution: [Write a CFG for the above scenario also derive the Sample Output [up to 2 rows only] from your CFG using Parse Tree]

CFG:

S → < table > R0 </table > S | < table > R0 </table >
R0 → < tr > C0 </tr > R1 | < tr > C0 </tr >
R1 → < tr > C1 </tr > R2 | < tr > C1 </tr >
R2 → < tr > C1 </tr > R0 | < tr > C1 </tr >
C0 → < td > Num </td > C0 | < td > Num </td >
C1 → < td > Num </td > C1 | < td > Num </td >
C1 → < td > Alpha </td > C1 | < td > Alpha </td >

Num → 0|1|2|3|4|5|6|7|8|9|0
Alpha → a|b| ... |z|A|B ... |Z

Parse Tree:

CLO 4 [10 + 14 =24 Marks]

Question 4:

a) Suppose you have a single tape Turing machine (TM) that is infinite from both ends. The current head (pointer) is at # (in the start of TM). Dry run the TM on next page and give the content of the tape after running it (When TM halts). Also mention the location of the head **[8.5+1.5 = 10 Marks]**

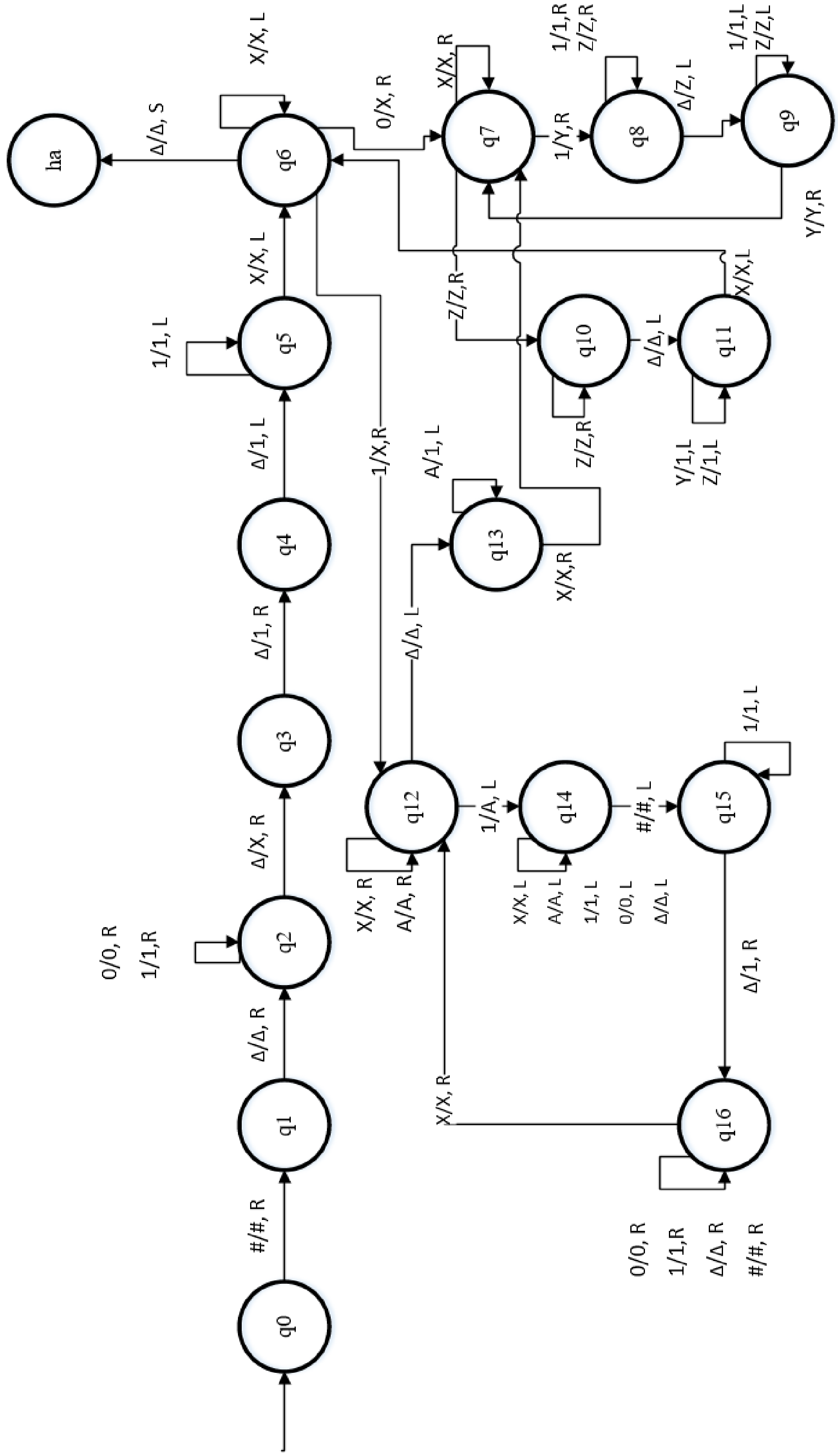
Input String:

.	.	.	Δ	Δ	#	Δ	**1**	**0**	Δ	Δ	Δ	.	.	.

Final Output along with head/pointer Location

ΔΔ1111#ΔXXX11111111ΔΔ

pointer: Δ after hash



q0
#/#, R
q1
Δ/Δ, R
q2
0/0, R
1/1,R
Δ/X, R
q3
Δ/1, R
q4
Δ/1, L
q5
1/1, L
X/X, L
q6
X/X, L
Δ/Δ, S
ha
0/X, R
q7
X/X, R
1/Y,R
q8
1/1,R
Z/Z,R
Δ/Z, L
q9
1/1,L
Z/Z,L
Y/Y,R
Z/Z,R
q10
Z/Z,R
Δ/Δ, L
q11
Y/1,L
Z/1,L
X/X,L
1/X,R
q12
X/X, R
A/A, R
Δ/Δ, L
q13
A/1, L
X/X,R
1/A, L
q14
X/X, L
A/A, L
1/1, L
0/0, L
Δ/Δ, L
#/#, L
q15
1/1, L
Δ/1, R
q16
0/0, R
1/1,R
Δ/Δ, R
#/#, R
X/X, R

b) Design a Multi-Tape Turing Machine that has two inputs X and Y. Both inputs are unary numbers. X is placed on tape 1 while Y is on tape 2. You need to perform X mod Y and store its remainder and quotient in Tape 3 and Tape 4 respectively. Sample input and output is given below. **[14 Marks]**

Note: X and Y are greater than 0.

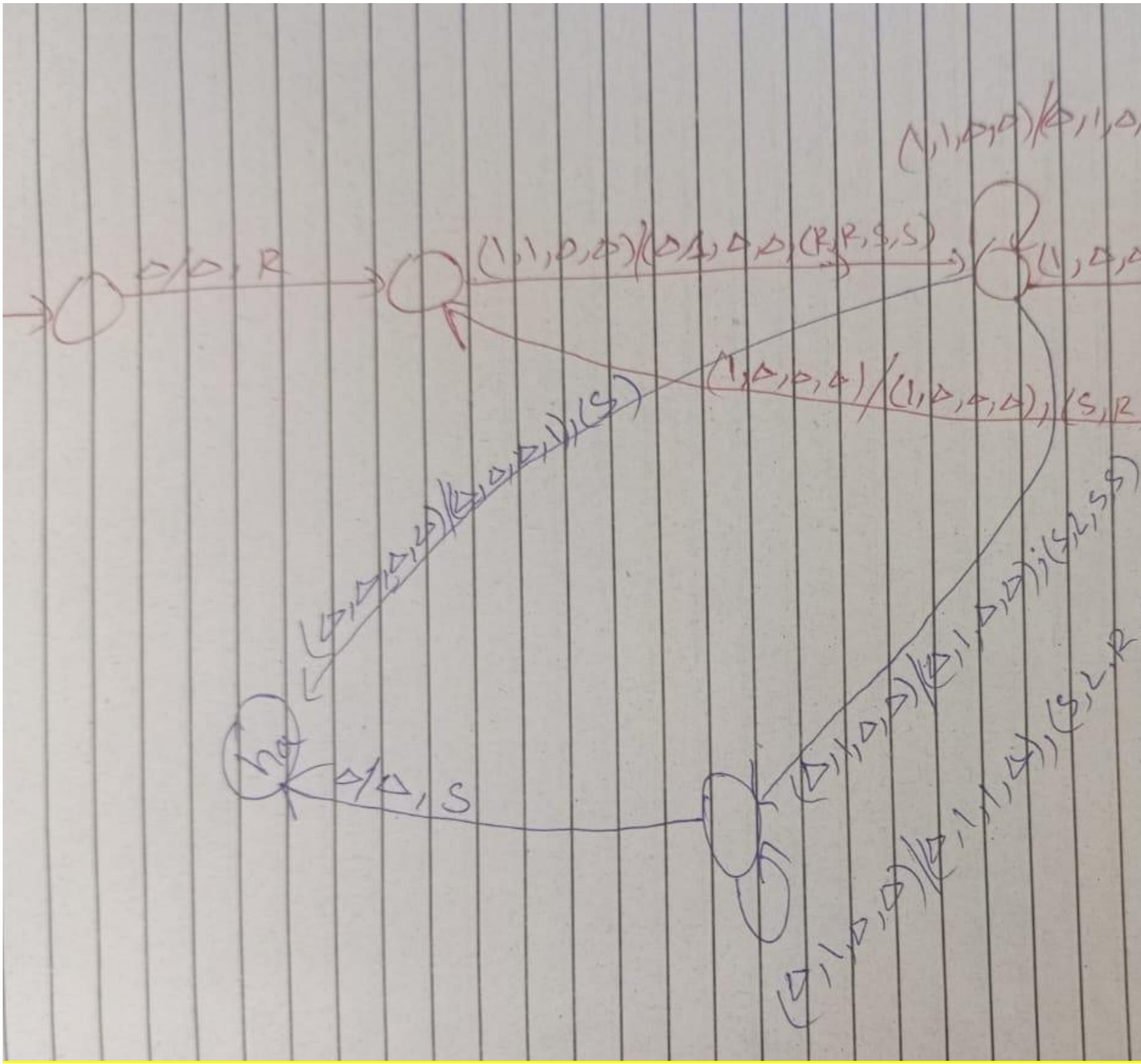
Sample Input:

Tape1	Δ	1	1	1	1	1	Δ
Tape2	Δ	1	1	1	Δ	Δ	Δ
Tape3	Δ	Δ	Δ	Δ	Δ	Δ	Δ
Tape4	Δ	Δ	Δ	Δ	Δ	Δ	Δ

Sample Output:

Tape1	Δ	1	1	1	1	1	Δ
Tape2	Δ	1	1	1	Δ	Δ	Δ
Tape3	Δ	1	1	Δ	Δ	Δ	Δ
Tape4	Δ	1	Δ	Δ	Δ	Δ	Δ

Turing Machine


Δ/Δ, R
ha
Δ/Δ, S

Rough Work

National University of Computer and Emerging Sciences
Lahore Campus

Theory of Automata (CS3005)

Final Exam

Date: May 30th 2024

Course Instructor(s)

Mr. Fraz Yousaf

Total Time (Hrs): 3

Total Marks: 65

Total Questions: 4

Roll No

Student Signature

Do not write below this line.

Attempt all the questions.

CLO #1: Identify formal language classes and prove language membership properties.

Question :1 [3 + 2 + 8]

a) If L1, L2 and L3 are Regular languages and L4 = L1 ∩ (L2 U L3). What kind of language will be L4. (RL, CFL or non-CFL) Explain briefly.

b) **True/ False** Can a DFA recognize a palindrome number?

c) *Tell whether the following Language is context free (CFL) or non- context free (non- CFL). If it is CFL provide PDA else prove it using Pumping Lemma.*

$$L = \{x \in \{a, b, c\}^* \mid n_a(x) < n_b(x) \text{ and } n_a(x) < n_c(x)\}$$

CLO #2: Differentiate and manipulate formal descriptions of languages, automata and grammars with focus on non-regular and regular using automata (DFA, NFA, NFA-Null)

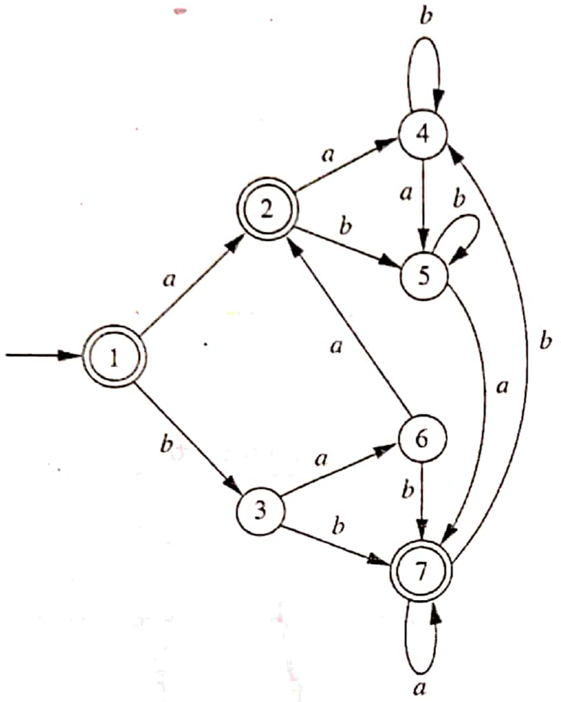
Question: 2 [2 + 7.5 + 7.5]

a) Which of the following is true?

a) (01)*0 = 0(10)*
b) (0+1)*0(0+1)*1(0+1) = (0+1)*01(0+1)*
c) (0+1)*01(0+1)*+1*0* = (0+1)*
d) All of the mentioned

b) Minimization of DFA

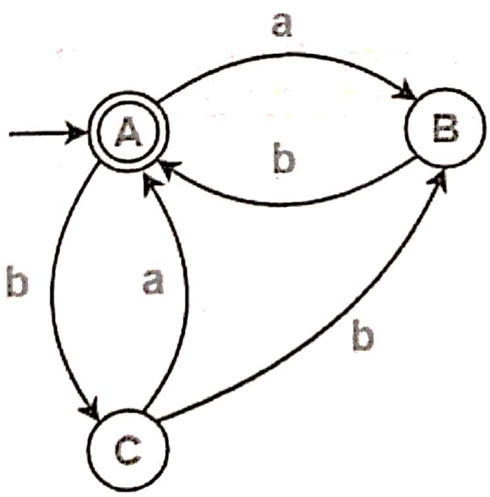
Find a minimum-state DFA recognizing the same language. Show complete working. Use only the method discussed in your respective class.



	a	b.
1	2	3
2	4	5
3	6	7
4	5	4
5	7	5
6	2	7
7	7	4

c) State Elimination Method

Use the State Elimination Method for extraction of Regular Expression. Write Final Regular Expression. Delete the states in increasing order of alphabets[First A then B then C]



CLO #3: Differentiate and manipulate formal descriptions of languages, automata, and grammars. with focus on context-free languages using automata (PDA and NPDA).

Question 3: [3 +5 +12 + 5]

a) Tell whether the following grammar is ambiguous. How? Give justification.

$S \rightarrow A \mid B$
$A \rightarrow aAb \mid ab$
$B \rightarrow abB \mid \epsilon$

b) Give a context-free Grammar for the regular expression $0^{*}\ 1(0+1)^{*}$

c) Dry run the single-tape Turing machine on next page and give the content of the tape after running it (When TM halts).

The initial configuration of the TM is given below.

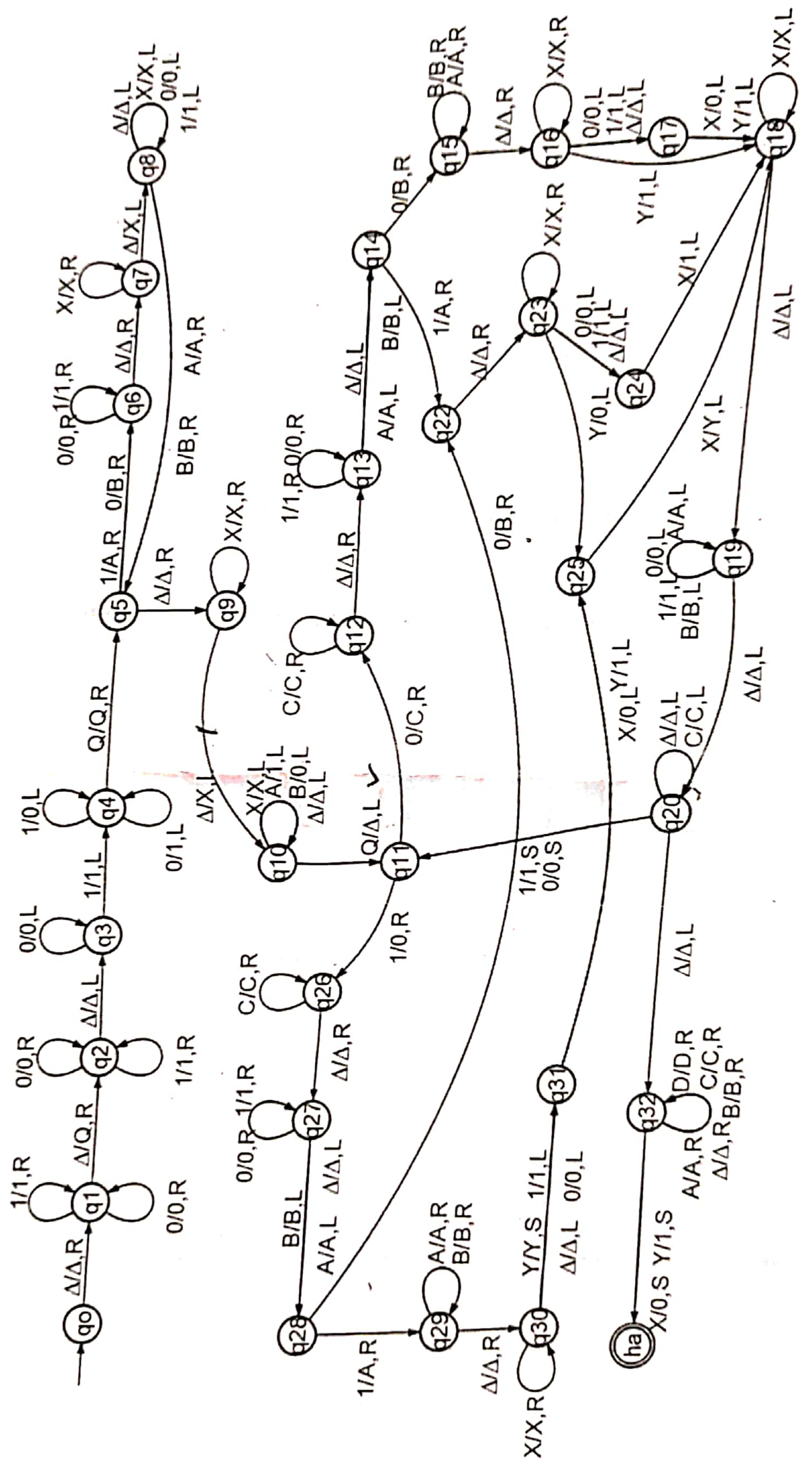
Δ	1	1	1	1	Δ	0	1	1	1	Δ	Δ	-----	Δ

⇧ head/pointer

Answer: You need to answer on the Answer Sheet.

Δ	0	0																	

Clearly show where will be the head/pointer when TM halts


qo
q1
q2
q3
q4
q5
q6
q7
q8
q9
q10
q11
q12
q13
q14
q15
q16
q17
q18
q19
q20
q22
q23
q24
q25
q26
q27
q28
q29
q30
q31
q32
ha
Δ/Δ,R
1/1,R
0/0,R
Δ/Q,R
Δ/Δ,L
0/0,L
1/1,L
1/0,L
0/1,L
Q/Q,R
1/A,R
0/B,R
X/X,R
Δ/X,L
X/X,L
B/B,R
A/A,R
A/1,L
B/0,L
Q/Δ,L
C/C,R
0/C,R
1/1,S
0/0,S
1/0,R
A/A,L
B/B,L
Y/Y,S
X/0,L
Y/1,L
X/1,L
Y/0,L
X/Y,L
C/C,L
D/D,R
C/C,R
B/B,R
X/0,S
Y/1,S

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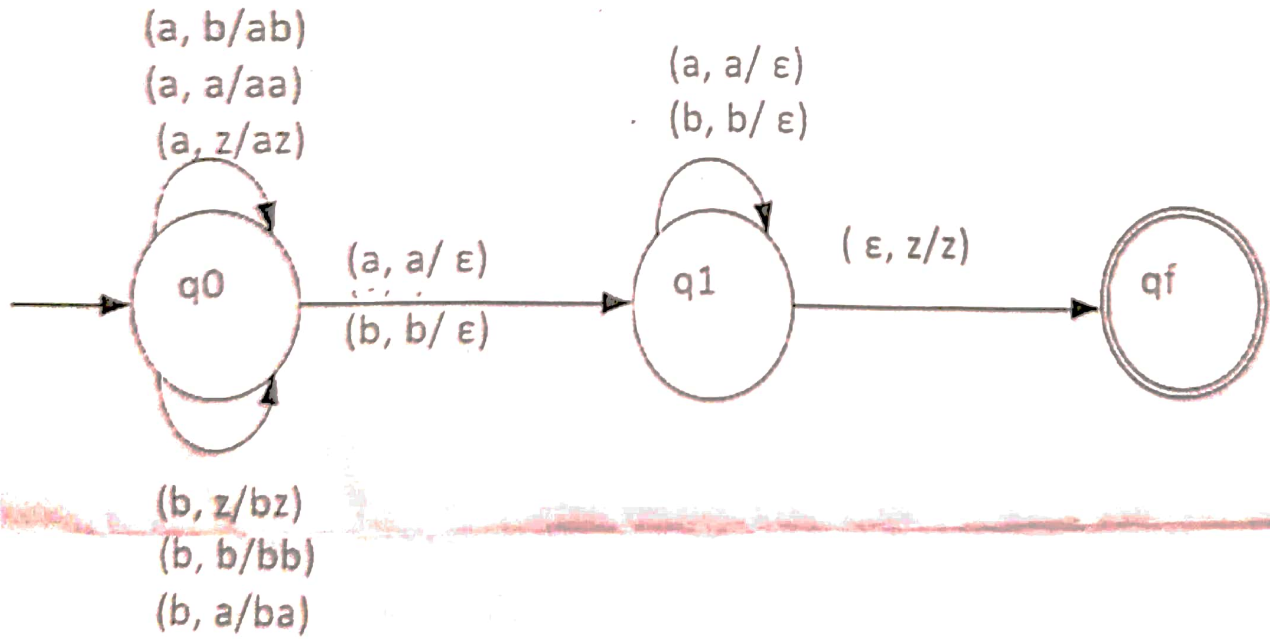
Lahore Campus

d) Initially, the stack contains the symbol "z". Design the parse tree for the string "aabbaa" using a non-deterministic pushdown automaton (NPDA), following the pattern discussed in class



CLO #4: Differentiate and manipulate formal descriptions of languages, automata and grammars with focus on non context-free languages using Turing Machines

Question: 4 **[10]**

Design a single-tape Turing machine to reverse the elements on the tape. Your machine has limited memory, NOT allowing you to replace any blank symbol with another symbol. You can modify each cell at most twice, ensuring all changes occur on the tape itself, except for blank symbols.

Sample Input 1:

Δ	1	1	1	0	1	0	Δ

Sample Output 1:

Δ	0	1	0	1	1	1	Δ

Sample Input 2:

Δ	0	0	0	1	1	1	Δ

Sample Output 2:

Δ	1	1	1	0	0	0	Δ

8010 / 0108

National University of Computer and Emerging Sciences
Lahore Campus

Theory of Automata (CS3005) Final- Exam

Date: Dec 31st 2024

Course Instructor(s)
Dr. Sobia Tariq Javed
Dr. Tahir Ejaz
Mr. Hamad ul Qudous
Mr. Fraz Yousaf

Total Time (Hrs): 3
Total Marks: 80
Total Questions: 4

Roll No ____ Section ____ Student Signature ____

Do not write below this line

- **Attempt all the questions. No marks without working.**
- **In case of confusion or ambiguity make a reasonable assumption.**
- **Solve Q3, Q4 and Q5 on the question paper. Show working on the answer sheet (MUST).**
- **Provide only one final solution to each question. Submitting multiple solutions will result in cancellation, and no marks will be awarded.** Attach question paper with answer book.

CLO	1	2		3	4		Total
Question#	Q1	Q3(a)	Q3(b)	Q4	Q2	Q5	
Marks (total)	10	10	10	10	20	20	80
Marks (obtained)							

CLO #1: Identify formal language classes and prove language membership properties.

Question#1 : Pumping Lemma **[marks 10]**

Tell whether the following Language is context-free (CFL) or non- context-free (non- CFL). If it is CFL provide PDA else prove it using Pumping Lemma.

$$L = \{a^{2^k} \mid k \geq 0\}$$

CLO #4: Differentiate and manipulate formal descriptions of languages, automata, and grammars with a focus on non-context-free languages using Turing Machines

Question #2: Multi-tape TM: String Left Rotation Check **[marks 20]**

Design a deterministic 2-tape TM to check if one string Y is a left rotation of another string X. You can assume X is on tape 1 and Y is on tape 2.
For example, "waterbottle" is a rotation of "erbottlewat". **X** = waterbottle and **Y** = erbottlewat
For your simplicity, we have reduced the input alphabet set to Σ= {a, b}

Accepting examples

Example # 1

Δ	**a**	**b**	**b**	**a**	**b**	**Δ**
Δ	**b**	**b**	**a**	**b**	**a**	**Δ**

Example # 2

Δ	**a**	**b**	**b**	**a**	**b**	**Δ**
Δ	**a**	**b**	**a**	**b**	**b**	**Δ**

Non-accepting examples

Example # 3

Δ	**a**	**b**	**b**	**a**	**b**	**Δ**
Δ	**b**	**b**	**a**	**b**	**b**	**Δ**

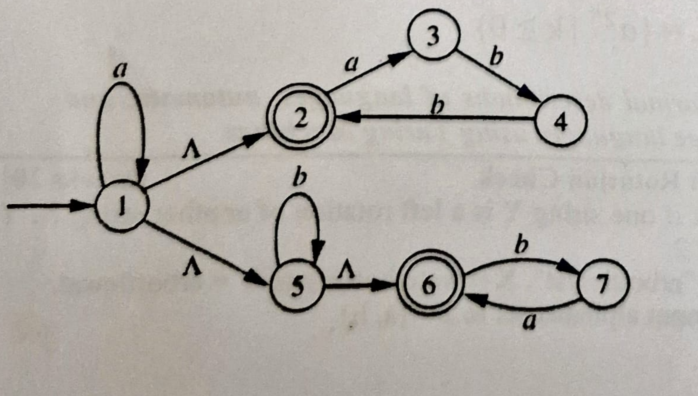
Example # 4

Δ	**a**	**b**	**b**	**a**	**b**	**Δ**
Δ	**a**	**b**	**a**	**a**	**b**	**Δ**

CLO #2: Differentiate and manipulate formal descriptions of languages, automata and grammars with focus on non-regular and regular using automata (DFA, NFA, NFA-Null)

Question#3 (a): NFA-NULL to NFA Conversion **[marks 10]**

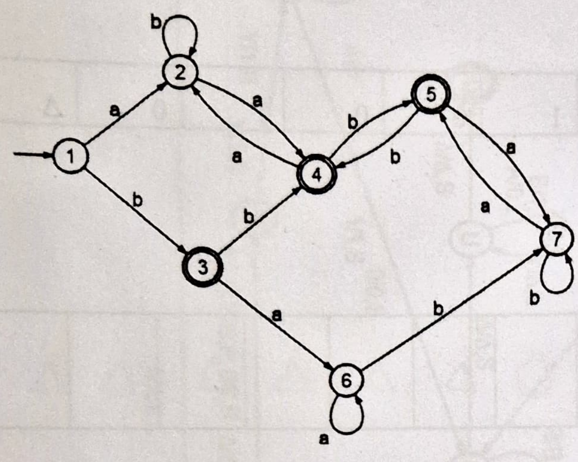
Convert the following **NFA-NULL** into an equivalent **NFA** by eliminating all Null transitions. Show your work step-by-step (on the answer sheet), including the computation of Null-closures for each state and the new transition table for NFA (fill in the table on the question paper).



States q	**δ (q,a)**	**δ (q,a)**
1		
2		
3		
4		
5		
6		
7		

Question#3 (b): Minimization of DFA **[marks 10]**

Find a minimum-state DFA recognizing the same language. Show complete working. Use only the method discussed in your respective class. Show working on answer booklet.



Minimized DFA

CLO #3: Differentiate and manipulate formal descriptions of languages, automata, and grammar. with a focus on context-free languages using automata (PDA and NPDA).

Question#4: CYK **[marks 10]**

For the following grammar, use the CYK algorithm to determine whether the string ***"abccba"*** is in the language of the grammar.

S→ AB | BB
A→ BA | c | b | CC
B→ BB | a | b | c
C→ AA

a b c c b a

					B
				AB	SB
			AB	ABCS	SB
		AB	ABCS	ABCS	SB
	AB	ABCS	ABCS	ABCS	SB
B	ASB	ABCS	ABCS	ABCS	SB

Circle the correct option:

Acceptable / Unacceptable

CLO #4: Differentiate and manipulate formal descriptions of languages, automata, and grammars with a focus on non-context-free languages using Turing Machines

Question#5: Dry Run Turing Machine **[marks 20]**

Perform a dry run of the single-tape Turing machine on the next page and provide the tape's final content after the machine halts. Additionally, describe the machine's function or purpose.

The initial configuration of the TM is given below.

\$	0	0	1	Δ	0	0	1	0	0	1	0	Δ

⇧ head/pointer

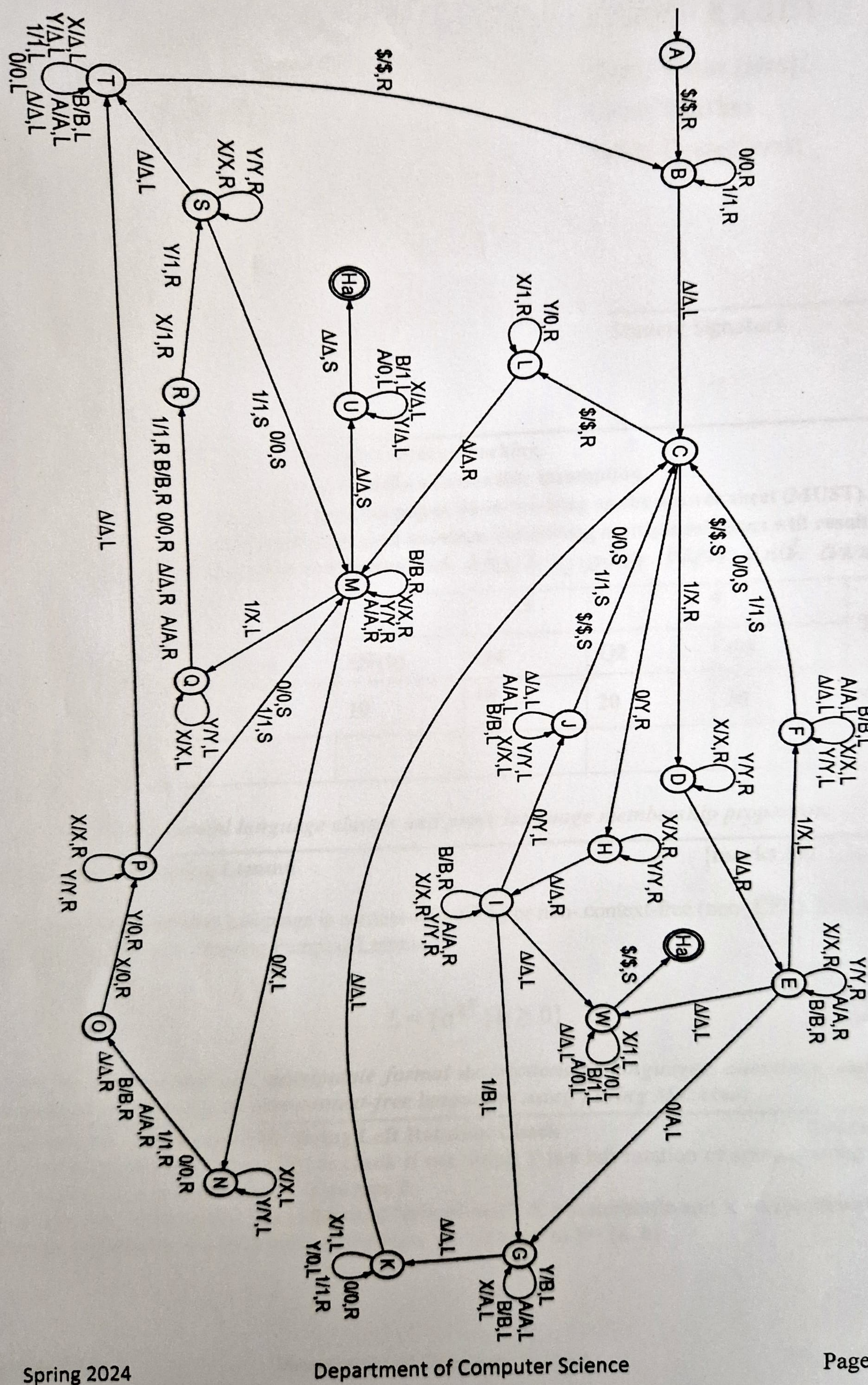
Final tape's content:

\$	0	0	1	Δ	0	0	1	0	Δ	Δ	Δ	Δ	Δ	Δ

↑

Also mark where will be the head pointer.

Working (be brief and to the point)


A
B
C
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U
W
Ha
Ha
$/$,R
0/0,R
1/1,R
Δ/Δ,L
$/$,R
Y/0,R
X/1,R
Δ/Δ,R
$/$,S
0/0,S
1/1,S
1/X,R
0/0,S
1/1,S
$/$,S
0/Y,R
0/Y,L
1/X,L
Δ/Δ,R
Δ/Δ,L
$/$,S
1/B,L
0/A,L
Δ/Δ,L
Δ/Δ,S
Δ/Δ,S
Δ/Δ,L
0/X,L
1/X,L
Y/1,R
X/1,R
Δ/Δ,L
Δ/Δ,L
Y/0,R
X/0,R
$/$,R

National University of Computer and Emerging Sciences

Lahore Campus

Theory of Automata (CS3005)

Final Exam **Solution**

Date: June 5th 2025

Course Instructor(s)

Dr. Tahir Ejaz

Total Time (Hrs): 2.5

Total Marks: 44

Total Questions: 4

____________ Roll No   ____________ Section   ________________________ Student Signature

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1. Rough sheets are *allowed* but you are **required to write solution in the provided space**.
2. ***Do not*** submit the rough sheets.

CLO # 3: *Differentiate and manipulate formal descriptions of languages, automata and grammars with focus on context-free languages using automata (PDA and NPDA).*

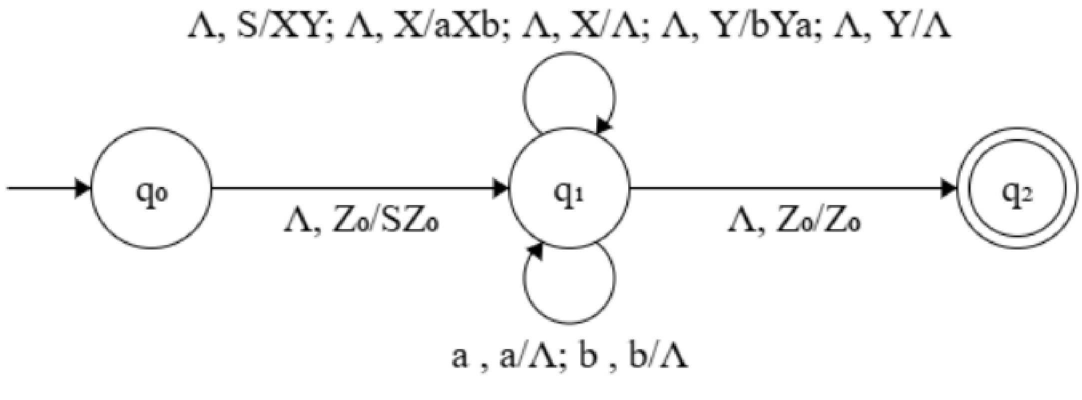
Q1: Draw the transition diagram of a *Pushdown Automata (not necessarily deterministic)* for the following language:

$$\{a^n b^{m+n} a^m \mid n, m \geq 0\}$$

Note: The number of states of your PDA must not exceed 3.

Hint: Every CFG has a corresponding 3 state PDA.

[15 Marks]



X generates $a^n b^n$, and Y generates $b^m a^m$; hence, XY generates $a^n b^{n+m} a^m$.

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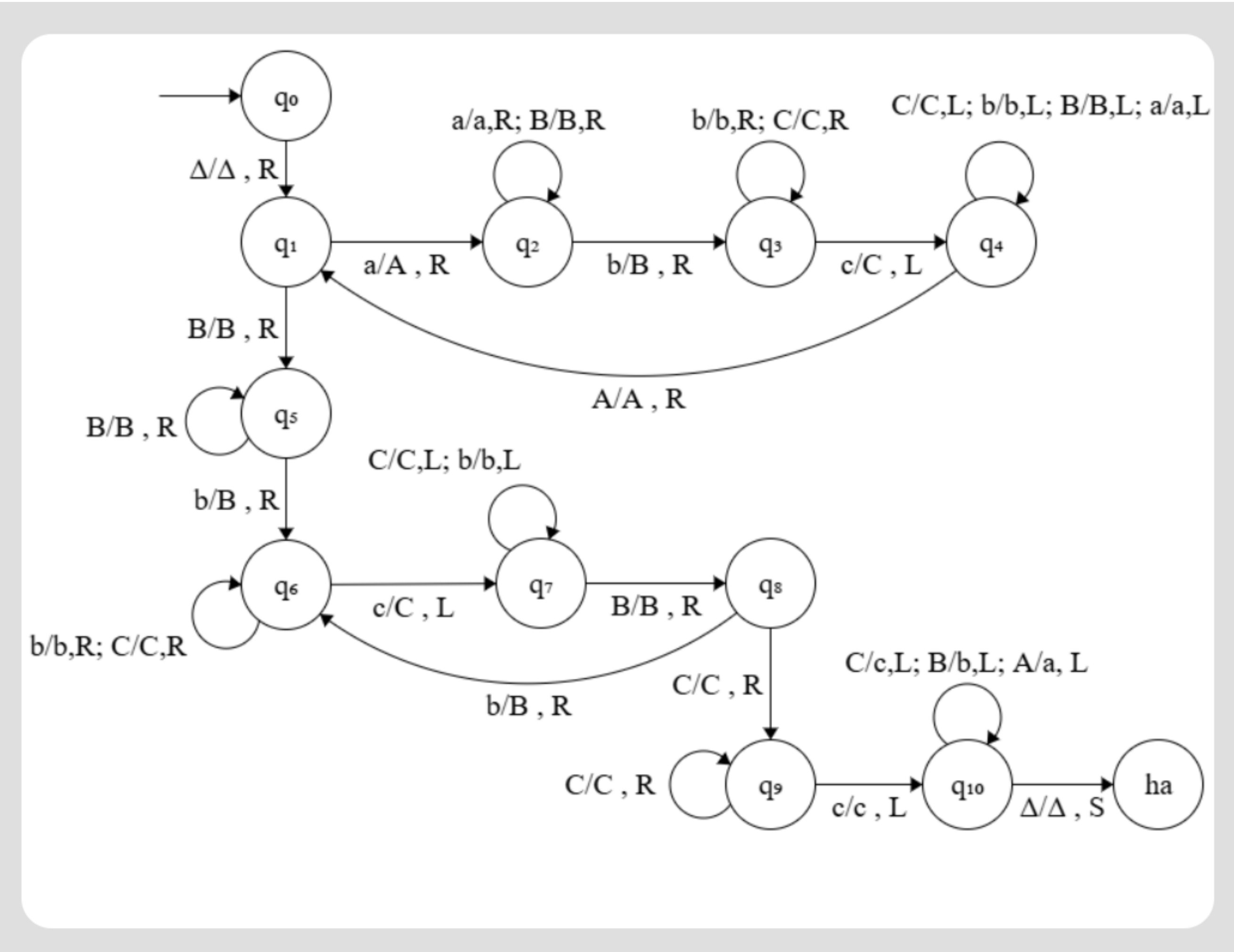
CLO # 4: *Differentiate and manipulate formal descriptions of languages, automata and grammars with focus on non context-free languages using Turing Machines*

Q2: Draw a transition diagram for a (single-tape, deterministic) Turing machine that accepts the following language:

$$L = \{a^i b^j c^k \mid 0 < i < j < k\}$$

Your machine must ***not*** enter in an **infinite loop**.
Note: For the sake of simplicity, assume that the input has the form $aa^*bb^*cc^*$.

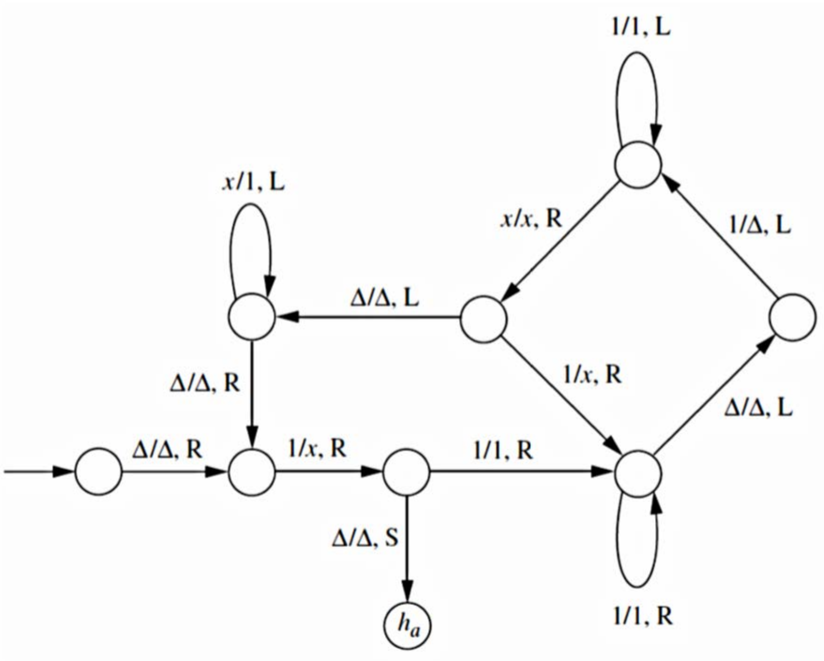
[15 Marks]



***CLO # 4:** Differentiate and manipulate formal descriptions of languages, automata and grammars with focus on non context-free languages using Turing Machines*

Q3: Describe the language (a subset of $\{1\}^*$) accepted by the TM in the following Figure:

[7 Marks]



Answer:
$L = \{w \in \{1\}^* \text{ such that } \lvert w \rvert \text{ is a power of } 2\}$

***CLO # 1:** Identify formal language classes and prove language membership properties. Prove and disprove theorems establishing key properties of formal languages and automata*

Q4: Using pumping lemma, we want to show that the following language L is not a regular language.

$$L = \{a^i b^j c^j \mid i \geq 1 \text{ and } j \geq 0\}$$

Let the pumping lemma constant be n (i.e. let's assume that there is an FA comprising of n states that accepts L); then what string should we choose to arrive at a contradiction? *Informally* describe why your chosen string should work.

[7 Marks]

Answer: ab^nc^n
***Justification:** If v contains a, then $uv^0w \notin L$; else, v must consist of b's only, and uv^2w has more b's than c's.*